Endogenous Model Uncertainty and Curiosity under Cognitive Limits

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October 2026

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Abstract

What stimulates curiosity: the drive to seek the information required to explore new perspectives? Neuroscience suggests that people automatically classify a situation before choosing how to act, drawing a familiar interpretation from similar past situations (e.g., a farmer attributing a neighbor's high yield to good weather). As observations become harder to reconcile with that interpretation, people increasingly entertain alternatives. This paper treats such shifts in "mindsets" as a choice of how much model uncertainty to entertain. Each mental model pairs a prior with a view of how observable features combine to reveal the state. After an initial signal, a decision maker with a costless default and latent, less precise alternatives jointly chooses model weights and how many subsequent signals to acquire. The key discipline is an evidence constraint that bounds motivated beliefs: an alternative's weight is capped by its Bayesian posterior. Curiosity then arises endogenously as the value of resolving the model uncertainty the decision maker chooses to entertain. Two results follow. First, the evidence constraint interacts with a complementarity between model weights and information acquisition to produce curiosity failures. Second, because decision makers can seek information and reason beyond what associative memory supplies, the framework isolates the influence of the information environment from hallmark behavioral effects in associative-memory models (e.g., Bordalo et al., 2025, 2026). Last, I endogenize the formation of mental models, formalizing creative and critical thinking as complementary forms of cognitive dimension reduction that sustain curiosity. I illustrate the theory throughout with development programs, teacher training, and agricultural technology adoption.

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A mind is not a vessel to be filled, but a fire to be kindled.

— Plutarch, *On Listening to Lectures*

The efficiency of a human problem solver depends critically on his ability to find a representation of the problem that restricts search to a small subspace of the total problem space.

— Allen Newell and Herbert A. Simon, *Human Problem Solving* (1972)

1 Introduction

Is seeing a doctor a path to better health, or a risk of the mistreatment once inflicted on people of one's race (Alsan and Wanamaker, 2018)? Is a neighbor's high yield the result of new farming practices, or simply of favorable rainfall? Is school-based agricultural education in rural Africa a way to build students' independent farming skills and scientific capabilities while spreading improved techniques to their households, or does it simply train students as farm labor at the expense of their studies, as many school gardens do (Lee, 2026)?

In each of these cases, how a person classifies the situation shapes her behavior decisively. Each classification tells her not only what to expect but where to look: a farmer who credits the rainfall has no reason to compare her neighbor's plots with her own. Importantly, when past experience leads a person to dismiss a promising opportunity, providing facts has limited effect, because whether her reading is correct becomes clear only through personal, repeated observations: repeated visits to the doctor, comparisons across plots and seasons, or visiting school farms and assigning trial plots to students, which few rural parents do regularly. Gathering this evidence takes time and resources, keeping track of an unfamiliar reading takes mental effort, and when past experience makes people confident in their familiar interpretation, such evidence promises too little to justify these costs. The focal question thus becomes how to stimulate curiosity—the drive for people to seek, on their own, the information required to explore new perspectives.

In fact, these examples reflect a general feature of cognition. A large body of neuroscience, most prominently cognitive map theory (O'Keefe and Dostrovsky, 1971; O'Keefe and Nadel, 1978; Hafting et al., 2005), suggests that people do not weigh every interpretation of a situation afresh. To economize on mental effort under cognitive limits, the brain instead assigns each situation to a familiar mental model[1] retrieved from memory of similar past experience, before deliberation begins (Behrens et al., 2018; Sanders et al., 2020). As observations become harder to reconcile with that model, it increasingly entertains alternative mental models, but there is also room for choice: the brain tends to favor mental models that promise higher payoffs if true and ignore those that are plausible but less rewarding (Shenhav et al., 2013; Lieder and Griffiths, 2020).

Allowing people to seek information and reason beyond what associative memory supplies, this paper models these contextual shifts in "mindsets" as a choice of how much model uncertainty to entertain after an initial signal—such as how a development program is presented. These "mindsets", which are competing models of the world formally known as mental models, are defined as a prior over the payoff-relevant state together with a likelihood that says how observable features combine to reveal that state. Two mental models therefore differ not only in what they expect but in what they attend to and how they read it: a farmer who credits the rainfall tracks the weather, and one who credits the seed tracks the plots. What each attends to

[1] I use the terms "mental models" and "cognitive maps" interchangeably. Neuroscientists speak of cognitive maps because the concept grew out of studies of how the hippocampus represents physical space, and was later extended to the organization of abstract knowledge (O'Keefe and Nadel, 1978; Behrens et al., 2018); psychologists use "mental models" for the internal representations people reason with. This paper relies only on what the two share: a structured representation that tells a person which observable features matter and how they combine to reveal an unobserved state. It makes no claim about how that representation is implemented in the brain, so the two terms can be used interchangeably here.

is not chosen feature by feature (as in models of selective attention, e.g., Schwartzstein, 2014; Hanna et al., 2014) but follows from the explanation each gives of how the state is produced. The decision maker is not restricted to holding only one mental model at a time, and weighting mental models changes how a person interprets a situation. The decision maker starts from a default mental model that is costless to use and faces a menu of latent alternatives whose priors are less precise than the default's and that yield expected payoffs that differ from those under the default. After observing the initial signal, she jointly chooses (i) the weight of attention she pays to each alternative and (ii) how many subsequent signals to acquire.

Regarding the choice of mental model weights, the framework integrates two decision theories under cognitive realism. As in Bayesian model averaging (e.g., Brock, Durlauf, and West, 2003; Cogley and Sargent, 2005), the posterior disciplines the weights, but the decision maker chooses them rather than adopting them: tracking an unfamiliar reading takes mental effort, and holding it forces a single action to hedge across models that recommend different actions, so she may place less weight on an alternative than its posterior. As in models of motivated beliefs (e.g., Brunnermeier and Parker, 2005; Bénabou and Tirole, 2016), expected payoffs guide that choice. However, I impose a key discipline from neuroscience: because the brain activates a mental model according to how well it predicts what is observed relative to its rivals, an evidence constraint caps the weight on an alternative at the posterior probability the evidence alone assigns it: the decision maker can neglect evidence but cannot invent it. This matches the psychology of motivated reasoning, in which people do not believe whatever they wish but construct justifications a dispassionate observer could accept, so that strong enough disconfirming evidence moves their beliefs (Kunda, 1990; Redlawsk et al., 2010).

In summary, the decision maker places weight on an alternative to the extent that acting on it promises, if it is right, a higher expected payoff than acting on the default, net of the costs of holding it, subject to the evidence cap. Much like collateral constraints in models of poverty traps (e.g., Galor and Zeira, 1993; Banerjee and Newman, 1993), the cap generates threshold effects in the exploration of alternative mental models. Consider an alternative that is more promising than the default—less promising ones are never activated. If it is right and evidence kept arriving, the cap would rise steadily and the weight would follow. Two features of the setup break this convergence: the initial signal fixes the evidence cap, and model weights and information are complements, since information is worth more under the vaguer alternative than under the precise default, so the more weight she places on the alternative, the more she seeks. Together they create a threshold. A weak initial signal caps the weight below the level at which investigating the alternative pays, so the decision maker never investigates it, the cap never rises, and the alternative stays unexplored—a curiosity failure in which she never seeks the information that could change her mind. A signal just above that level starts a self-reinforcing cycle in which the weight justifies information, the information raises the cap, and the higher cap justifies more weight, so small differences in the initial signal produce sharply different outcomes. Parents in Africa may therefore ignore a school-based agricultural education program that fails to distinguish itself from typical school gardens that use students as farm labor, yet a light-touch information intervention can prompt them to seek information on their own, through school visits, trials of improved techniques, and delegating farms to students (Lee, 2026).

Characterizing the choice of model uncertainty allows curiosity to be defined formally, as an endogenous object rather than a fixed trait. Once an alternative carries weight, the decision maker must act without knowing which mental model is right, which forces her to hedge and makes resolving the difference valuable. Curiosity is defined as the value of resolving that model uncertainty, a function of model weights: zero under the default alone, greatest when the mental models are most balanced, and falling away as either prevails, matching the well-known finding that curiosity peaks at intermediate confidence (Loewenstein, 1994; Kang et al., 2009). Adding ambiguity aversion, a concave aggregator over the weights she chooses, integrates noninstrumental motives of curiosity: it raises both the cost of holding an alternative and the value of resolving which mental model is right, with a premium that grows with the gap in expected payoffs between the mental models. High-stakes questions—such as whether fertilizer bought on credit will double a farm's income or, if it is fake, leave the family in debt (Bold et al., 2017; Michelson et al., 2021)—are therefore harder to open

but, once open, more actively pursued, so interventions that activate alternative mental models become more effective relative to those that make information cheaper to acquire.

Curiosity also manifests in the type of information the decision maker seeks, once her exposure to signals is made endogenous. The choice of signal trades off learning more within the mental models she holds (refinement) against learning which of them is right (resolution), on terms set by the weight she places on the alternative. With little model uncertainty she seeks few signals, and those the default reads; as model uncertainty grows, curiosity directs her toward the features on which the mental models disagree, most strongly when the answer would change what she does; once the alternative dominates, she seeks signals the alternative reads. Shifts in "mindsets" are thus reflected in what information the decision maker seeks: a farmer who credits the rainfall watches the weather; one who suspects a new practice compares plots; one who has adopted it studies how to apply it.

What, then, do we gain from modeling humans' capacity to be curious, on top of a default classification based on associative memory? My framework revises two hallmark behavioral effects in associative-memory models. The first is context effects: adding payoff-irrelevant cues changes which representation of the situation comes to mind and so changes behavior (Bordalo et al., 2026); a consumer sees a jar of jam in plain packaging as a routine purchase and focuses on its price, but the same jar in fancy packaging becomes a treat and draws her attention to its quality. In my framework, a cue is an observable feature of the initial signal and counts as evidence like any other: it shifts weight only toward a mental model that predicts it better than the others, so cues sway behavior only if they are payoff-relevant under some mental model. When a cue activates such an alternative, the curiosity this triggers tests it: with unbiased evidence the shift persists if the alternative fits better than the default and reverses otherwise, and when one of the two mental models is true, a tendency to seek confirming evidence only slows the reversal. Persistent context effects therefore point to evidence that is scarce, costly, or biased, or to a menu with no true mental model, rather than to a fixed cognitive bias. Because people have an endogenous drive to seek this evidence themselves, policy can make it accessible instead of delivering it: for farmers who distrust fertilizer that looks clumped but meets nutrient standards (Michelson et al., 2021), the remedy is quality testing they can obtain on their own.

Thus far, the discussion has pitted a single alternative against the default, but practical questions often admit many explanations: a farmer who sees a neighbor's high yield may credit the weather, the soil, a new seed variety, fertilizer, or the timing of planting. My framework allows many alternatives, and because the brain's automatic matching of cues against stored mental models weighs all of them at once, the support the evidence lends one shrinks with every rival that also fits it. Interference effects, the second hallmark, therefore also work differently. In associative-memory models, experiences compete for retrieval and the representation with the greatest cue similarity comes to mind (Bordalo et al., 2025). Here, rivals that fit the same evidence divide its support, so several alternatives that each outperform the default can all fail to be activated, and even a cue that lends plausibility to an alternative the decision maker would never entertain can stifle exploration of a promising one.

Interference effects thus raise a question that a fixed menu of mental models cannot answer: how does a mind with limited capacity keep its menu small enough to explore? I next endogenize the menu and introduce creative and critical thinking as two complementary forms of cognitive dimension reduction. Both rest on a premise from neuroscience: mental models are assembled from reusable pieces of explanation—formally known as templates—each a causal mechanism linking an underlying cause to observable features, such as rain to wet soil and high yields (Pearl, 2009; Behrens et al., 2018; Whittington et al., 2020), and the brain imagines new situations by recombining stored experience (Hassabis et al., 2007; Schacter and Addis, 2007).

As with curiosity, creative thinking is evidence-based: the decision maker builds new mental models from templates she already holds, and a new one is admitted only if the evidence favors it over the one it would replace. Vertical creative thinking adds a template to an existing mental model, explaining the same situation more deeply, such as crediting a neighbor's yield to the soil as well as the rain; it is admitted when what the current mental model fails to predict is surprising enough along the new template's direction. Horizontal

creative thinking extends a mental model to new situations, such as carrying a lesson learned on cassava over to rice; it is admitted when the situations depart from expectations in the same direction. I prove that the two are not independent, yet neither substitutes for the other: a template that explains why a technique works sets how far a mental model can transfer, since reach is fixed by the templates it contains, but each transfer remains a horizontal step the decision maker must take and the evidence must justify, and such a template is worth nothing in a single situation, paying only through transfer. Creative thinking is thus not a fixed trait but bounded by the templates the decision maker holds, so providing templates expands what she can create and how far it transfers. For training teachers and agricultural extension officers, this qualifies a common presumption that teaching why a practice works is enough to make trainees generalize. Each transfer is a creative step the trainee must take and test in the new classroom or farm, and it happens only if her curiosity clears the threshold there; training in mechanisms therefore pays through her continued curiosity and horizontal creative thinking, not in place of them. This complements recent work valuing conceptual knowledge within a single decision (Davies and Sankar, 2026) and showing that mechanistic explanations help farmers generalize (Sankar et al., 2026): the framework specifies when that value is realized across situations and when it is not.

Creative thinking has a potential pitfall, however: a mental model assembled after seeing the evidence can fit it by chance. Critical thinking, a standing habit of mind, guards against this: it means not using the same data twice, so that every mental model, new or old, is judged only by evidence not used to form it. I model it as a parameter $\kappa \in [0,1]$, how strongly the decision maker discounts an explanation found after the fact for the breadth of the search that produced it, as in multiple hypothesis testing. I prove that a threshold value of κ, strictly below one and rising with the templates' vagueness, separates two regimes: below it, the probability of a false insight converges to one as the number of candidates searched grows, so discounting is useless; above it, that probability converges to zero. Critical thinking has its own potential pitfall, and it is a curiosity failure rather than a rejection: the discount can hold a correct explanation below the threshold at which the decision maker would seek the evidence bearing on it, and more so the broader the search that found it. Without complementary triggers for creative thinking, school-based agricultural education programs that cultivate critical thinking face a tradeoff between fewer false insights and slower adoption of correct ones: students trained to think critically should be less likely to adopt a new planting date or seed variety that merely coincided with a good harvest, but slower to adopt effective crop spacing they discover themselves.

Taken together, the results suggest that open-mindedness, curiosity, creative thinking, and critical thinking form a capability growth cycle within the brain's evidence-based reasoning. Open-mindedness—entertaining an alternative—starts curiosity, but curiosity tends to extinguish itself as the evidence it gathers resolves the uncertainty that prompted it. Creative thinking can replenish that uncertainty, since the surprises curiosity uncovers justify new mental models, while critical thinking judges each against fresh evidence and keeps chance fits off the menu, where they would dilute the evidence cap of the right alternative. These capabilities are often treated as general traits or skills that a person carries across settings (McCrae, 1987; Halpern, 1998; Heckman and Kautz, 2012; Kashdan et al., 2018). Viewed through the brain's evidence-based reasoning, however, even a general capability must start somewhere: the evidence from a particular situation sets it in motion, and carrying it to new situations is a creative step that the evidence must justify and that costs mental effort. This offers one reading of why efforts to train these capabilities in the abstract and in isolation have shown limited transfer across domains (Willingham, 2007; Sala and Gobet, 2017).

The same problem arises for training developing-country teachers in new pedagogies that go beyond transmission and retention of subject content. Such pedagogies ask teachers to connect knowledge to practical needs, which differ across classrooms and shift quickly as developing economies transform, so training must enable teachers to adapt what they learn to diverse challenges and keep it relevant as conditions change. How, then, can they learn to do so and keep growing their capabilities? Since teaching why a practice works does not by itself make trainees generalize, the framework suggests integrating two ingredients to keep the capability growth cycle going. The first is training that, rather than handing out recipes—mental models

that may be too rigid to deploy widely—targets the fundamental templates underlying the new pedagogy, from which teachers assemble mental models suited to their own classrooms: reflection that sets new mental models against old ones sharpens the templates, and group practice and simulation with peers let teachers see the resulting mental models work until they are confident enough to explore their consequences in their own classrooms. The second is a stream of opportunities: because a shift to a new mental model must promise enough to pay for itself, opportunities that raise the returns to the new pedagogy and make evidence on them accessible keep teachers building and testing new mental models as their communities change. Section 7 discusses my ongoing research that tests this idea, integrating a highly effective "Learning to Learn" approach to teacher training (Nourani et al., 2025, 2026) with school-based agricultural education as a platform that offers experiential science learning opportunities and high-return farming techniques (Lee, 2026).

Organization. Section 2 discusses the model setup and key assumptions. Section 3 characterizes alternative activation, curiosity failures, the impact of the initial signal, a formal definition of curiosity, and where the decision maker seeks information. Section 4 illustrates how behavioral effects are revised relative to associative-memory models. Sections 5 and 6 endogenize the menu of mental models through creative and critical thinking. Section 7 offers concluding remarks.

2 Setup

A decision maker (DM) faces an unknown payoff-relevant state and holds a menu of mental models of her situation: a costless default and K latent alternatives. The structure follows cognitive map theory, which originally dealt with spatial navigation but has recently been extended to learning about conceptual ideas, memory mapping, and social relationships (Constantinescu et al., 2016; Behrens et al., 2018; Whittington et al., 2020). The default is a familiar preconception, more precise than the alternatives and free to run as an "auto-pilot" mode (Kool et al., 2010; Shenhav et al., 2013); activating an alternative requires cognitive effort, and holding several at once costs more than holding each alone (Shenhav et al., 2017; Lieder and Griffiths, 2020). After an initial observation, the DM chooses jointly (i) the attention weights she places on the alternatives, which set how much model uncertainty she entertains, and (ii) how many subsequent costly signals to acquire, that is, "active" learning. She then takes a single act that must serve every model she holds. Besides the cognitive cost, an evidence constraint disciplines the weights: no alternative may receive more weight than the evidence supports. The DM maximizes expected utility; ambiguity aversion enters only as an extension, where it adds a non-instrumental motive for curiosity and strengthens the results. The menu is taken as fixed until Sections 5 and 6, which endogenize its formation through a primitive called templates.

2.1 Mental models, signals, and timing

Let $\theta \in \mathbb{R}$ denote an unknown payoff-relevant state: the return to a course of action whose value the DM does not know. The DM has von Neumann–Morgenstern preferences represented by $u(a, \theta)$ over acts $a \in \mathscr{A} \subseteq \mathbb{R}$ and states θ, and $\mathbb{E}_\mu\, u(a, \theta)$ denotes expected utility under a belief μ.

Initial signal. The DM observes an initial signal, a raw feature vector $x \in \mathbb{R}^d$. For example, what a parent sees or hears of a school-based agricultural education program includes what students do on the school farm, which crops and techniques are in use, how far students are involved in management decisions, and whether the program is integrated with experiential science instruction. The initial signal is exogenous to the DM but can be designed by an intervention: a promotional video that pitches the distinguishing features of the program, together with the program itself, would be one such x.

Mental models. The DM holds a finite menu of mental models of the situation she faces, indexed by $k \in \{0, 1, \ldots, K\}$. Index $k = 0$ is the *default* model, which she has applied many times before and which she operates on costlessly; indices $k = 1, \ldots, K$ are *latent alternatives*, which she may or may not bring to bear.[2] Until Sections 5 and 6, the menu is fixed: subsequent signals reweight and refine the models on it rather than introducing new ones. Taking it as given is where an associative account of memory enters: what a DM can entertain is what the situation has brought to mind, so retrieval determines the menu and the problem studied here begins once retrieval has happened. A mental model has two aspects. The first is a *prior* over the state: model k asserts $\theta \sim \mathcal{N}(\mu_k, \sigma_k^2)$ with precision $\tau_k \equiv 1/\sigma_k^2$, and the default is the most precise on the menu, $\tau_0 > \tau_k$ for every $k \geq 1$, because it has been applied most often. The means are left unordered. The second aspect is a *generative process* that says how the observable features combine to reveal the state. This is the model's explanation of how the situation produces what is observed, and it determines both which features the model attends to and how it interprets them. As the cognitive-map theory treats it, each model is a complete generative process for the features of the initial signal:

$$\theta \sim \mathcal{N}(\mu_k, \sigma_k^2), \qquad x \mid \theta \sim \mathcal{N}(\theta\, b_k,\, \Sigma), \qquad \tau_k := 1/\sigma_k^2, \qquad k = 0, 1, \ldots, K, \tag{1}$$

where $b_k \in \mathbb{R}^d \setminus \{0\}$ is the mental model's loading vector and $\Sigma \succ 0$ is a common noise covariance. Each alternative is structurally distinct from the default and from the other alternatives, in that it carries its own loading b_k.

Each mental model also implies an *attention mapping*: the weights it places on the components of the initial signal when it forms its belief about θ. The mapping is not an extra primitive; it is what each mental model's own inference problem implies. Features a model does not encode are not learned from, however long the experience. Hanna et al. (2014) document such a failure among seaweed farmers who never learned from years of their own harvests that pod size mattered. In their model a farmer can keep track of only a few of the many inputs that affect yield, and pod size was not among the ones she tracked. Here a farmer does not choose which inputs to track one by one. Each mental model is an explanation of how yields are produced, and that explanation determines which features she looks at: a farmer who credits the rainfall tracks the weather, and one who credits the seed tracks the plots. Attention changes when the mental model does.

Lemma 1 (Attention mappings are derived)**.** *Under* (1)*, the posterior over θ given x depends on x only through the scalar*

$$s_k \;=\; \alpha_k' x, \qquad \alpha_k \;=\; \frac{\Sigma^{-1} b_k}{b_k' \Sigma^{-1} b_k}, \tag{2}$$

and $s_k \mid \theta \sim \mathcal{N}(\theta, \nu_k^2)$ with $\nu_k^2 = (b_k' \Sigma^{-1} b_k)^{-1}$. The posterior is Gaussian with precision $P_k(0) = \tau_k + \nu_k^{-2}$ and mean $m_k = \big(\tau_k \mu_k + \nu_k^{-2} s_k\big)/P_k(0)$.

Each mental model thus compresses the observable features into a single number s_k carrying everything it needs to know about θ, and the weights in that compression, α_k, are its attention mapping: the combination of features the model treats as informative. Attention is thus a consequence of the explanation the model gives, not a separate choice over features, and it changes when the model does. I assume throughout that the default remains the most precise model after reading the initial signal, $P_0(0) > P_k(0)$ for every $k \geq 1$, so that every alternative has more left to learn than the default.[3] When $b_j \neq b_k$ two models read the same raw observation differently, and features diagnostic under one may be ignored by the other, so shifting weight between them

[2] A latent alternative need not come from experience; it is available but carries no weight until activated. Latent structure of this kind has a neural counterpart: sequences that later form the mental model of a new environment are already present in the hippocampus before that environment is first experienced (Dragoi and Tonegawa, 2011). Sections 5 and 6 discuss in detail how mental models are formed from templates and how the resulting menu is pruned.

[3] Substantively, this assumption means that a situation has one clear default rather than several close contenders: a single familiar model that repeated experience has made sharply more precise than the rest, which is how cognitive map theory describes assignment, with the brain settling on one well-learned map for a context rather than holding several in balance (Wills et al., 2005; Sanders et al., 2020).

changes both what the DM looks at and how she interprets the same situation; when $b_j \propto b_k$ they attend identically and differ only in (μ_k, τ_k). The distinction has observable content, since models with different loadings assign different likelihoods to the same features. This gives a formal edge to what is otherwise verbal: two priors differing only in their means are *not* two mental models in any sense that matters here.

Subsequent signals. After observing the initial signal x, the DM may acquire $n \geq 0$ subsequent signals at marginal cost c per signal, each of the form $y_i = \theta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \nu_y^2)$.

Attention weights. The DM chooses a vector of attention weights $\mathbf{w} = (w_1, \ldots, w_K)$ on the alternatives,[4] with $w_k \geq 0$ and $\sum_{k\geq 1} w_k \leq 1$, the remainder $w_0 = 1 - \sum_{k\geq 1} w_k$ resting on the default; and she chooses the number of subsequent signals n. Timing is:

(1) The initial signal x is realized and observed.

(2) The DM chooses the weights w_k on the alternatives, each bounded above by an evidence constraint defined below, and the number of subsequent signals $n \in \{0, 1, 2, \ldots\}$.

(3) The n subsequent signals realize.

(4) The DM chooses a single act a, evaluated at the weights chosen in step (2).

The results concern the weights chosen at step (2). Step (4) fixes the structure of the problem. The DM chooses a single act under model uncertainty, so the maximization runs over one a evaluated at the aggregated weights. Aggregating model-contingent maxima $\max_a \mathbb{E}_k u(a, \theta)$ would instead evaluate the position of a DM who *knows* which mental model applies. Placing the maximum outside the sum over models is what gives the disagreement between mental models its bite, and it generates the hedging cost defined below.

The model is a single choice followed by a Bayesian update: the DM chooses $(\mathbf{w}, n)$ once, the signals realize, the weights update, and she acts. Where the analysis follows the DM over several rounds of information, it repeats this stage, with the updated bounds carried forward to the next stage and the DM choosing myopically in each.[5]

2.2 Preferences, costs, and the objective function

The DM evaluates an act by its expected utility under each mental model and aggregates across models with the weights. To nest attitudes toward model uncertainty, the aggregation is written through a strictly increasing $\varphi : \mathbb{R} \to \mathbb{R}$, as in the smooth ambiguity model (Klibanoff et al., 2005), except that the weights are chosen rather than given by preferences. The baseline sets $\varphi(U) = U$: the DM maximizes expected utility under the mixture of models and has no ambiguity aversion as such. A strictly concave φ would add ambiguity aversion, as discussed in Section 3.5.

Formally, τ_0 reflects the many past situations in which the default was applied, whereas ν_k^{-2} is the information a single observation carries, so the assumption says that one observation does not undo the advantage that repeated experience has built; it holds whenever the loadings are equally strong, $\nu_k = \nu_0$, and fails only if an alternative's loading is far stronger than the default's.

[4] When the context is ambiguous, the brain does not commit to a single mental model (Sanders et al., 2020). Instead, it flickers between familiar and new mental models and searches for evidence until it settles on one. This flickering lasts fractions of a second in spatial navigation (Jezek et al., 2011), longer in the perception of ambiguous figures (Leopold and Logothetis, 1999), and plausibly months for the conceptual mental models studied here. A decision taken after deliberation integrates the mental models the brain has visited.

[5] Myopia rules out buying signals today for the weight they would license tomorrow, the experimentation motive of bandit models. Such planning requires representing an alternative well enough to value what learning it would permit, the deliberate effort the model treats as costly, and the neuroscience describes reassignment as ongoing inference rather than planned exploration (Sanders et al., 2020). Allowing it would lower the thresholds below for alternatives that promise enough.

Holding alternatives active imposes a cognitive cost,

$$C(\mathbf{w}) \;=\; \frac{\gamma}{2}\sum_{k\geq 1} w_k^2 \;+\; \eta \sum_{1\leq j<k\leq K} w_j w_k, \qquad \eta > \gamma > 0. \tag{3}$$

The cost has three properties. First, operating under the default alone is costless. Second, it is convex: each additional unit of attention to an alternative requires progressively greater inhibition of the dominant default and progressively greater suppression of interference from other active alternatives.[6] Third, because $\eta > \gamma$ the cost is superadditive across alternatives: holding two alternatives at once costs strictly more than the sum of holding each alone. When a single alternative k is held, (3) is simply $(\gamma/2)w_k^2$. The DM weighs the value function against this cost and the information-acquisition cost cn, selecting $\mathbf{w}$ and n to maximize the objective below subject to $n \in \mathbb{Z}_{\geq 0}$ and to an evidence constraint on each w_k defined below.

Writing $\mathbb{E}_{k,n}$ for the expectation under mental model k's posterior after x and n signals, and $\mathbb{E}_y$ for the expectation over signal realizations taken at step (2), the derived objective is

$$V(\mathbf{w}, n; x) \;=\; \mathbb{E}_y\Big[\max_{a\in\mathscr{A}} \sum_{k=0}^{K} w_k\, \varphi\big(\mathbb{E}_{k,n}\, u(a,\theta)\big)\Big] \;-\; C(\mathbf{w}) \;-\; cn. \tag{4}$$

2.3 The evidence constraint

The DM maximizes (4) subject to an evidence constraint: the weight on an alternative cannot exceed the evidentiary support for that alternative.[7] Let $L_k(x)$ be the marginal likelihood of the initial signal under mental model k and $\Lambda_k(x) := L_k(x)/L_0(x)$ the Bayes factor of alternative k against the default. Define

$$\overline{W}_k(x) \;:=\; \frac{L_k(x)}{L_0(x) + \sum_{j=1}^{K} L_j(x)} \;=\; \frac{\Lambda_k(x)}{1 + \sum_{j=1}^{K} \Lambda_j(x)}, \qquad k = 1, \ldots, K. \tag{5}$$

The right-hand side is the posterior probability that a Bayesian with a uniform prior over the $K+1$ mental models would assign to alternative k after seeing x, so the constraint $w_k \leq \overline{W}_k(x)$ reads: *the DM may not be more convinced of an alternative than the evidence justifies*. When x is consistent with the default, every $\overline{W}_k \approx 0$; when x falls in the default's tails, the bounds of the alternatives that account for x rise.

Two limiting cases. The constraint places the DM between two decision theories. As in Bayesian model averaging (Brock, Durlauf, and West, 2003; Cogley and Sargent, 2005), the posterior disciplines the weights, but the DM chooses them rather than adopting them: were holding models costless, she would set every weight at its cap and nothing would be chosen. As in models of motivated beliefs (Brunnermeier and Parker, 2005; Bénabou and Tirole, 2016), expected payoffs guide that choice: were there no cap, she would set the weights by the value of the models they favor, tempered only by the costs. The evidence constraint keeps her between the two: she may place less weight on an alternative than its posterior but never more, so she can neglect evidence but cannot invent it. This is the discipline neuroscience finds in how the brain activates a mental

[6] The convex form is the natural reduced-form summary of two well-documented neural constraints. Working-memory capacity is severely limited, to roughly three to four items (Cowan, 2001), and the precision with which items are held drops nonlinearly rather than linearly as memory load increases (Bays and Husain, 2008). And tracking competing explanations is more demanding than tracking items, because each hypothesis carries its own predictions that must be updated against incoming evidence.

[7] The constraint has a neural basis. Hippocampal remapping behaves like Bayesian inference over which context applies: the mental model that becomes active is the one favored by the posterior over contexts given the evidence (Sanders et al., 2020), so categorical activation is justified by evidence rather than preference. Taking that automatic assignment as $p_k(x) = \pi_k L_k(x)/\sum_{j=0}^{K} \pi_j L_j(x)$, with π the prior it carries, deliberation can hold a model only to the extent the assignment has made it available, $w_k \leq p_k(x)$; with a uniform π this is (5), and a default-favoring π rescales L_0. A well-learned mental model is an attractor that snaps back after a brief disturbance (Wills et al., 2005; Jezek et al., 2011), which is the one-sidedness below. The constraint is thus cognitive in origin though evidentiary in content; with the cognitive cost (3) it forms the cognitive limits of the title.

model, according to how well it predicts what is observed relative to its rivals (Sanders et al., 2020; Wills et al., 2005; Jezek et al., 2011), and psychology finds in motivated reasoning: instead of unconstrained beliefs, people construct justifications that a dispassionate observer could accept, and strong enough disconfirming evidence moves their beliefs (Kunda, 1990; Redlawsk et al., 2010).[8]

2.4 Bayesian updating

Every signal triggers two types of Bayesian updating. Within each active mental model, the posterior over θ updates by Bayes' rule, which is automatic and refines precision. Across mental models, the same signal is evidence about which of them generated it, and the DM uses it to reassess $\mathbf{w}$. The evidence caps (5) discipline the second type.

Write $\Lambda_k^y(y^n)$ for the likelihood ratio of the n subsequent signals under alternative k against the default, each mental model evaluated at its own step-(2) posterior. After the signals the weights and the evidence caps update by Bayes' rule,

$$w_{k,n} \;=\; \frac{w_k\,\Lambda_k^y(y^n)}{w_0+\sum_{j=1}^K w_j\,\Lambda_j^y(y^n)}, \qquad \overline{W}_{k,n} \;=\; \frac{\Lambda_k(x)\,\Lambda_k^y(y^n)}{1+\sum_{j=1}^K \Lambda_j(x)\,\Lambda_j^y(y^n)}, \tag{6}$$

and carry forward to the next stage; the within-model posteriors update as before. This is the recursive update of the second-order prior in Klibanoff et al. (2009), applied to weights the DM has chosen rather than inherited. Updating preserves the discipline: the odds of every alternative against the default never exceed its Bayes factor, and the weights are a martingale under the mixture, so acquiring signals cannot be expected to move attention in either direction, only to resolve it (Lemma OA.1 in the Online Appendix).[9]

2.5 The hedging cost of holding several mental models

A single act must serve whatever mixture of mental models the DM holds. The *hedging cost* of holding weights $\mathbf{w}$ is the value lost because the DM must choose one act rather than the act each mental model recommends:

$$A(\mathbf{w}) \;:=\; \underbrace{\sum_{k=0}^K w_k\,\varphi\big(U_k\big)}_{\text{value of knowing the mental model}} \;-\; \underbrace{\max_a \sum_{k=0}^K w_k\,\varphi\big(\mathbb{E}_k u(a,\theta)\big)}_{\text{value of one act under model uncertainty}}, \tag{7}$$

where $U_k := \max_a \mathbb{E}_k u(a,\theta)$ is the model-contingent value. The hedging cost is nonnegative, zero when all weight rests on one mental model or when every model receiving weight induces the same act, and concave in $\mathbf{w}$, since it is an affine function minus a maximum of affine functions.

[8] Given the weights, the DM's belief about θ is the $\mathbf{w}$-mixture of the mental models' posteriors, a single distribution. The two-model language is needed not to describe what she believes but what governs it: a Bayes factor exists only between identified components, the cognitive and hedging costs are functions of the weights, and the updating below tracks which component each piece of evidence favors. None of these survives the collapse to a single prior.

[9] Within a stage the weights move only by Bayes' rule and are a martingale under the mixture, so the DM anticipates them correctly and her choice of act is dynamically consistent; re-choosing the weights is deliberation and begins the next stage.

2.6 Specifying the optimization problem in closed form

The objective (4) is defined for general payoffs and signals. For tractability I specialize to a quadratic payoff and Gaussian signals.[10] Let the act be a scale of engagement $a \in \mathbb{R}$ with

$$u(a,\theta) \;=\; a(\theta-\chi) - \tfrac{\lambda}{2}a^2, \qquad \lambda > 0, \tag{8}$$

where χ is the marginal cost of engagement and λ an adjustment cost. Subsequent signals are conditionally i.i.d. draws $y_i = \theta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \nu_y^2)$, so mental model k's posterior precision after x and n signals is $P_k(n) = \tau_k + \nu_k^{-2} + n\nu_y^{-2}$, and the variance of its step-(4) posterior mean, viewed from step (2), is

$$\psi_k(n) \;=\; \frac{1}{P_k(0)} - \frac{1}{P_k(n)}, \qquad \psi_k' > 0, \quad \psi_k'' < 0. \tag{9}$$

The model-contingent value is then

$$U_k(x,n) \;=\; \frac{(m_k-\chi)^2 + \psi_k(n)}{2\lambda}, \qquad \frac{\partial U_k}{\partial n} \;=\; \frac{1}{2\lambda\,\nu_y^2 P_k(n)^2} \;>\; 0, \tag{10}$$

increasing and concave in n.

In the baseline $\varphi(U) = U$, and (7) evaluates to

$$A(\mathbf{w}) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(m)}{2\lambda} \;=\; \frac{1}{2\lambda}\sum_{k=0}^{K} w_k\,(m_k - \bar{m})^2, \qquad \bar{m} := \sum_{k=0}^{K} w_k m_k, \tag{11}$$

the weighted variance of the model-implied posterior means. When a single alternative k is held,

$$A \;=\; w_k(1-w_k)\,D_k(x), \qquad D_k(x) \;:=\; \frac{(m_k - m_0)^2}{2\lambda}, \tag{12}$$

where $D_k(x)$, the *disagreement* between alternative k and the default, measures how far apart the two mental models' recommended acts are given the signal; it is not a parameter of the DM, since each posterior mean reads the initial signal through its own attention mapping. The DM's problem becomes

$$\max_{\mathbf{w},\,n} V(\mathbf{w},n) \;=\; \sum_{k=0}^{K} w_k U_k(x,n) \;-\; A(\mathbf{w}) \;-\; C(\mathbf{w}) \;-\; cn \quad \text{s.t.} \quad 0 \le w_k \le \overline{W}_k(x),\; k \ge 1,\;\; n \in \mathbb{Z}_{\ge 0}, \tag{13}$$

and, when a single alternative k is held at weight w_k,

$$V(w_k,n) \;=\; (1-w_k)\,U_0(x,n) + w_k\,U_k(x,n) \;-\; w_k(1-w_k)D_k(x) \;-\; \tfrac{\gamma}{2}w_k^2 \;-\; cn. \tag{14}$$

Equation (14) makes the intuition explicit. Weight on an alternative offers returns in the value gap $U_k - U_0$, at the hedging cost $w_k(1-w_k)D_k$ and the cognitive cost $(\gamma/2)w_k^2$; signals refine each mental model's estimate through ψ_k and shrink the model uncertainty in expectation, at a price of c per signal. Since the hedging cost measures the model uncertainty the DM holds, problem (13) is the choice of how much model uncertainty to entertain, through the weights, and how much of it to resolve, through the signals.

[10] These functional forms do not affect the qualitative results: the evidence constraint, the crowding it produces, and the classification of regimes do not depend on them (Remark OA.1 in the Online Appendix). They allow activation conditions to be stated as explicit thresholds and closed forms, and provide the Gaussian structure in which the later results are stated.

3 Main Results

This section presents the main results of the paper, which characterize the conditions under which the DM entertains alternative mental models and seeks the information required to explore them. I first characterize the thresholds in alternative activation, the curiosity failures that arise when the evidence constraint meets a complementarity between model weights and information acquisition, and the initial signal required to clear the thresholds. I then show that curiosity can be defined endogenously as the value of resolving the model uncertainty that activation creates. Finally, I enrich these results by adding a non-instrumental motive for curiosity through ambiguity aversion and by endogenizing the type of information acquired.

A property of the cognitive cost function simplifies the results in this paper.

Lemma 2 (Spreading attention across alternatives is costly)**.** *Fix total attention off the default, $W = \sum_{k\geq 1} w_k$. Then $C(\mathbf{w})$ in* (3) *is uniquely minimized by placing all of W on a single alternative, and the cost of any interior split strictly exceeds the concentrated cost by $(\eta - \gamma)\sum_{j<k} w_j w_k > 0$.*

The DM's problem thus has a built-in preference for entertaining one alternative at a time, the endogenous counterpart of a restriction imposed by assumption elsewhere in the model-switching literature (Ortoleva, 2012; Schwartzstein and Sunderam, 2021). When a single alternative k is held, the objective is (14): a comparison between alternative k and the default, with the disagreement $D_k(x)$, the value gap $\Delta U_k := U_k(x,n) - U_0(x,n)$, and the evidence cap $\overline{W}_k(x)$ all computed from the initial signal. The DM entertains the alternative for which the optimized value of (14) is highest, and the initial signal determines it through the same three objects. The next subsections characterize activation for a given alternative before returning to the interaction between alternatives in Section 4.2.

3.1 Alternative activation

The DM places weight on an alternative to the extent that acting on it promises, if it is right, a higher expected payoff than acting on the default, net of the costs of holding it, subject to the evidence constraint.

Activation of alternative k is governed by a value condition and the evidence constraint. The *value condition* is that the value gap ΔU_k clear a threshold T_k, so that the alternative is worth enough to pay for holding both models.[11] The *evidence constraint* is that the weight on k cannot exceed $\overline{W}_k(x)$, its evidence cap, which is near zero when x is consistent with the default. Which form the value condition takes depends on the curvature of (14) in w_k, namely $2D_k - \gamma$, so T_k is stated regime by regime. Proposition 1 establishes two results: there is a threshold T_k only above which the value gap earns the alternative any weight, and the weight never exceeds the evidence cap; and the threshold takes two forms, a graded regime in which the weight rises continuously once it is crossed and an all-or-nothing regime in which it jumps to the cap.

Proposition 1 (Activation thresholds)**.** *Fix n and an alternative k held on its own, and write $D_k = D_k(x)$ and $\overline{W}_k = \overline{W}_k(x)$.*

(i) (Thresholds.) *There is a threshold $T_k \leq D_k$ such that $w_k^* = 0$ if $\Delta U_k \leq T_k$ and $w_k^* > 0$ if $\Delta U_k > T_k$, and in either case $w_k^* \leq \overline{W}_k$.*

(ii) (Two regimes.) *The threshold and the form of the optimum depend on the curvature $2D_k - \gamma$.*

(a) (Graded regime, $\gamma > 2D_k$.) *$T_k = D_k$, V is strictly concave in w_k, and*

$$w_k^* \;=\; \text{med}\Big\{\, 0,\; \frac{\Delta U_k - D_k}{\gamma - 2D_k},\; \overline{W}_k \,\Big\},$$

[11] "Worth" here means decision value, not optimism. Under (10), U_k rises with how decisive the action model k recommends is, $(m_k - \chi)^2$, and with what remains to be learned under it, $\psi_k(n)$. An alternative that reads the opportunity as harmful, and so calls for decisively opting out, can therefore carry a larger value gap than a lukewarm default. What the DM never activates is an alternative that would change her action too little, or teach her too little, to pay for holding it.

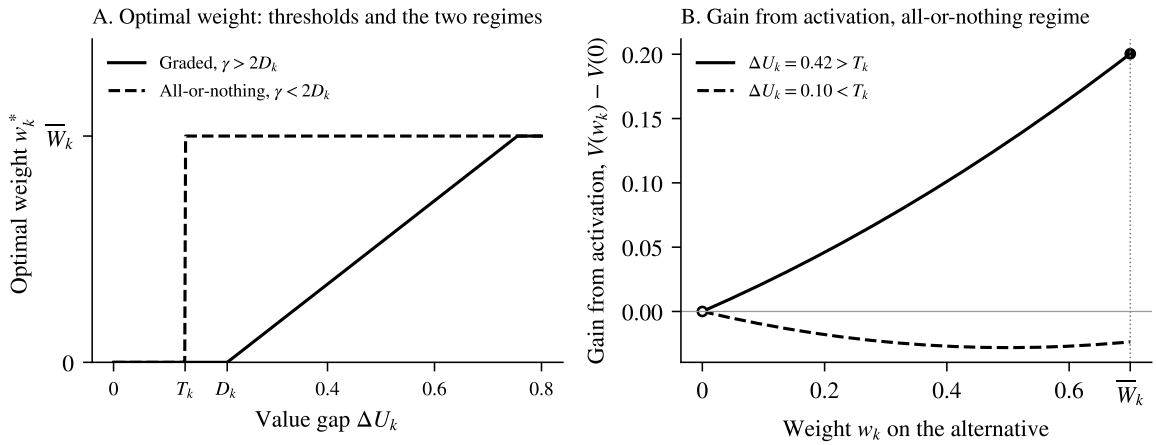


FIGURE 1. THRESHOLD EFFECTS IN ALTERNATIVE ACTIVATION

Notes: Both panels illustrate Proposition 1 with $D_k = 0.2125$ and evidence cap $\overline{W}_k = 0.70$. Panel A plots the optimal weight against the value gap in the two regimes. In both, the weight is zero until the value gap clears the threshold T_k and never exceeds the evidence cap (part (i)). The threshold is marked on the horizontal axis and differs across regimes: in the graded regime ($\gamma = 1.20 > 2D_k$) it equals the disagreement, $T_k = D_k = 0.2125$, and the weight rises continuously beyond it until it reaches the evidence cap; in the all-or-nothing regime ($\gamma = 0.20 < 2D_k$) it is lower, $T_k(\overline{W}_k) = D_k(1-\overline{W}_k) + (\gamma/2)\overline{W}_k = 0.134$, and the weight jumps from zero to the evidence cap there (part (ii)). Panel B plots the gain from activation, $V(w_k) - V(0)$ from (14), in the all-or-nothing regime. The gain is convex in the weight, so the optimum is a corner: for a value gap above $T_k(\overline{W}_k)$ (solid) the DM moves straight to the evidence cap, and for a value gap below it (dashed) any partial activation is worse than none.

which is continuous and increasing in ΔU_k.

(b) (All-or-nothing regime, $\gamma < 2D_k$.) *V is strictly convex in w_k, so $w_k^* \in \{0, \overline{W}_k\}$, and $w_k^* = \overline{W}_k$ if and only if*

$$\Delta U_k \;>\; T_k \;=\; T_k(\overline{W}_k) \;:=\; D_k\,(1-\overline{W}_k) + \tfrac{\gamma}{2}\overline{W}_k \;<\; D_k.$$

The threshold is strictly below D_k: when the alternative and the default disagree enough, a decisive shift can pay even though no partial shift would. The threshold T_k is strictly decreasing in $\overline{W}_k$, and at the threshold w_k^* jumps discontinuously from 0 to $\overline{W}_k$.

Both $\overline{W}_k(x)$ and $D_k(x)$ are functions of the initial signal, so the initial signal determines which alternatives can be activated and, for each, whether the weight on it moves continuously or in switches.

Figure 1 illustrates Proposition 1. Panel A asks whether the value gap exceeds T_k. Below it the alternative gets no weight; above it the weight is positive and bounded by the evidence cap $\overline{W}_k$. The threshold is the hedging cost of the first unit of weight: holding an alternative opens model uncertainty where there was none, at a marginal cost of D_k, so an alternative must be worth more than the disagreement it introduces, and one that disagrees sharply with the default is *harder* to activate partially. Above the threshold the weight rises continuously in the graded regime and jumps to the cap in the all-or-nothing regime. Panel B shows why it jumps. The curvature $2D_k - \gamma$ compares two marginal costs of weight: the hedging cost falls as the weight rises, because a mixture is most uncertain when most evenly split, while the cognitive cost rises. When disagreement is large relative to γ, compromise is the costlier way to hold the alternative and the DM resolves her model uncertainty by switching; when it is small, attention is the costlier way and she adjusts at the margin. The evidence constraint bounds the weight in both regimes but works differently in each. In the graded regime it binds only once the value gap is large enough, and the threshold stays at D_k. In the all-or-nothing regime it binds at every activated optimum, since the switch is to the cap, and it shapes the threshold: a higher cap lowers $T_k(\overline{W}_k)$, because the hedging cost per unit of weight, $(1-w_k)D_k$, falls as w_k rises toward 1, so a shift to a higher cap carries less model uncertainty per unit of weight.

The two regimes are the counterparts of the gradual and abrupt transitions neuroscientists observe when

the neurons encoding an animal's situation meet discrepant evidence: *rate remapping*, in which cells keep their firing fields and change only their rates (Leutgeb et al., 2005a), and *global remapping*, the all-at-once switch to an independent code (Wills et al., 2005; Leutgeb et al., 2005b; Colgin et al., 2008) that Sanders et al. (2020) interpret as Bayesian inference about which situation the animal is in.

3.2 Complementarity and curiosity failures

Proposition 1 holds the number of signals fixed. The DM chooses w_k and n together, and the two choices are complements: weight on an alternative raises the value of signals about it, and signals raise the value of weight. This subsection states the complementarity, derives the information acquisition it implies, and shows how, combined with the evidence constraint, it produces curiosity failures.

Proposition 2 (Complementarity)**.** *The effect of an additional signal on the return to attention to alternative k, the cross-partial of* (4)*, is*

$$\frac{\partial^2 V}{\partial w_k \, \partial n} \;=\; \frac{\partial U_k}{\partial n} - \frac{\partial U_0}{\partial n} \;=\; \frac{1}{2\lambda \nu_y^2}\Big[\frac{1}{P_k(n)^2} - \frac{1}{P_0(n)^2}\Big] \;>\; 0,$$

since the alternative is the less precise model, $P_k(n) < P_0(n)$. Hence (w_k, n) are complements: w_k^* is nondecreasing in n and n^* is nondecreasing in w_k, and the first-order condition in n is $\sum_{j=0}^{K} w_j \, \partial_n U_j = c$.

Weights and information are complements because the alternative has more left to learn: the marginal value of a signal falls with a model's precision, so weight on the less precise alternative raises the value of a signal, and a signal raises the value of weight. To see how the complementarity interacts with the evidence constraint, it helps to solve for the information the DM acquires in closed form, as a function of the weight she places on the alternative. The closed form is simplest when the subsequent signals bear on the alternative alone, so that information has value only through the weight on it, which the following assumption imposes.

Assumption 1 (Alternative-specific signals)**.** The subsequent signals load on features the default mental model does not encode: $\partial U_0 / \partial n = 0$.

Under Assumption 1 the marginal benefit of a signal at $\mathbf{w} = 0$ is zero, so $n^* = 0$ under the default alone. Information seeking is thus absent under a single mental model, however diffuse its prior, and grows with the attention the alternative receives: there is no point acquiring detail about a model that has not been activated.[12]

Corollary 1 (Closed-form information acquisition)**.** *Under Assumption 1, the optimal number of signals when alternative k is held on its own at weight w_k is*

$$n^*(w_k) \;=\; \nu_y^2 \Big[\Big(\frac{w_k}{2\lambda \nu_y^2 c}\Big)^{1/2} - P_k(0)\Big]^+,$$

which is zero for $w_k \le 2\lambda \nu_y^2 c \, P_k(0)^2$ and thereafter increasing and concave in w_k and decreasing in c. The optimum (w_k^*, n^*) is the fixed point of two explicit best responses: the weight $w_k^*(n)$ of Proposition 1, which depends on n through the value gap $\Delta U_k(n)$, and the number of signals $n^*(w_k)$ above.

Curiosity failures. Much like collateral constraints in models of poverty traps (Galor and Zeira, 1993; Banerjee and Newman, 1993), the evidence constraint generates threshold effects in the exploration of alternative mental models. Corollary 1 contains an information threshold: no signal is bought until the weight on

[12] The assumption is used for two things only: the result that no information is acquired under the default alone, $n^* = 0$ at $\mathbf{w} = 0$, and the closed form for the number of signals in Corollary 1. The complementarity holds without it (Proposition 2), and Section 3.6 drops it and recovers the same structure.

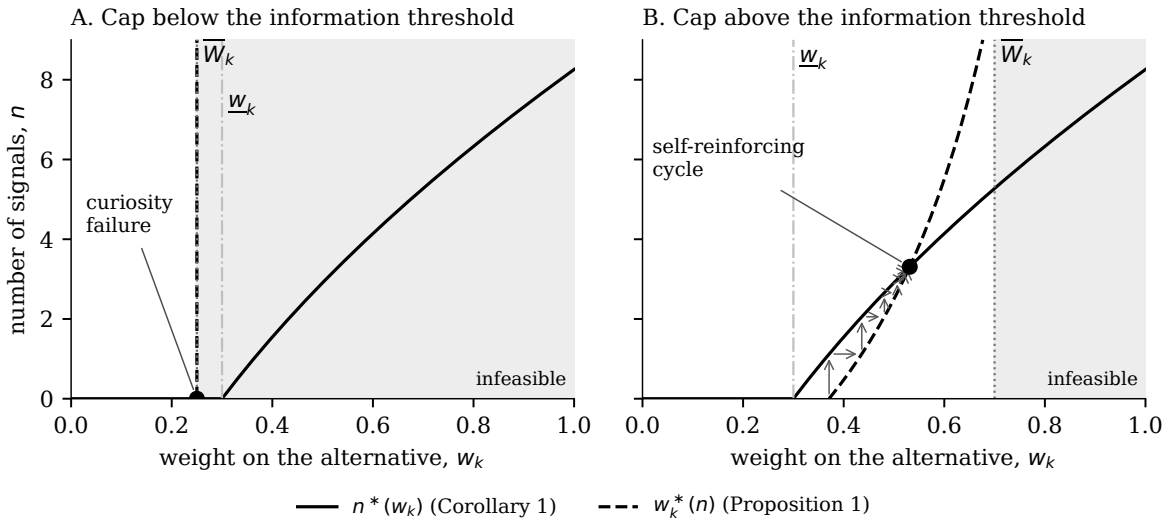


FIGURE 2. CURIOSITY FAILURES

Notes: Both panels plot, in the space of the weight on the alternative and the number of signals, the two best responses whose fixed point is the optimum of Corollary 1: the number of signals $n^*(w_k)$ given the weight (solid) and the weight $w_k^*(n)$ given the signals (dashed), in the graded regime of Proposition 1. Parameters are $\lambda = 1$, $\nu_y^2 = 10$, $P_k(0) = 1$, $c = 0.015$, $D_k = 0.2125$, $\gamma = 1.20$, and $\Delta U_k(0) = 0.50$, so the information threshold is $\underline{w}_k = 2\lambda\nu_y^2 c P_k(0)^2 = 0.30$; n is treated as continuous. The shaded area is infeasible under the evidence constraint. Panel A sets the cap at $\overline{W}_k = 0.25 < \underline{w}_k$: the weight is clamped at the cap, where no signal pays, so the dashed best response coincides with the cap line and the fixed point is $(0.25, 0)$. Panel B sets $\overline{W}_k = 0.70 > \underline{w}_k$: starting from the weight the DM would choose with no signals, the arrows trace the complementarity of Proposition 2, each round of signals justifying more weight and each increase in weight justifying more signals, converging to $(0.53, 3.3)$.

the alternative exceeds $\underline{w}_k := 2\lambda\nu_y^2 c P_k(0)^2$, a level that rises with the cost of signals and with the alternative's own precision. By Proposition 1, the weight cannot exceed the evidence cap $\overline{W}_k(x)$, and within a stage the initial signal fixes the cap. If $\overline{W}_k(x) \leq \underline{w}_k$, therefore, the alternative may be activated but is never investigated: $n^* = 0$, its posterior stays imprecise, and by (6) its cap never rises. This is a *curiosity failure*: the DM never seeks the information that could change her mind, however promising the alternative and even if it is right. If instead $\overline{W}_k(x) > \underline{w}_k$ and the alternative is activated at a weight above $\underline{w}_k$, the complementarity of Proposition 2 starts a self-reinforcing cycle: the weight justifies signals, the signals raise the cap through (6) when the alternative fits them, and at the next stage the higher cap admits more weight. Small differences in the initial signal can thus produce sharply different outcomes. Above the threshold, the path follows the regime of Proposition 1: in the graded regime the weight tracks the evidence continuously and signals are bought as it rises; in the all-or-nothing regime the weight switches to the cap and signals follow.

Figure 2 illustrates. Both panels plot the two best responses of Corollary 1 for the same alternative; only the evidence cap differs. In Panel A the cap lies below the information threshold $\underline{w}_k$, so the DM activates the alternative at the cap but buys no signal: the fixed point is $(\overline{W}_k, 0)$, a curiosity failure. In Panel B a more favorable initial signal lifts the cap above $\underline{w}_k$; the weight justifies signals and the signals justify weight, and the best responses meet at an interior fixed point with $n^* > 0$.

Lee (2026) provides a vivid example of curiosity failures. In a three-year experiment across 197 Liberian schools, a voluntary program trained science teachers to integrate inquiry-based learning with student-led school demonstration farms and home farming projects. Without further information, parents avoided the program for three years: only a third ever visited a school farm, school farm visits fell by 12% and parent–teacher meetings by 39%, technology adoption barely moved, and households pulled back from schooling, with absenteeism up 31% and studying hours down 20%. Where promotional videos and field days presented the program's distinguishing features, both responses reversed: parents sought information on their own via

school farm visits and parent–teacher meetings, their households adopted 26–53% more yield-improving techniques, dropout fell by 4–5 percentage points per year, absenteeism fell by 28%, and studying hours rose by 22%. What the two arms have in common is the parents' starting point: they read the program as a typical school garden that uses students as farm labor. That reading is the default of the theory, and a program that fails to distinguish itself from it leaves the evidence cap of the alternative reading, a program that builds students' skills and spreads techniques, below the level at which investigating it pays. The alternative is held at most faintly and never investigated, its cap never rises, and avoidance persists: a curiosity failure. The videos and field days change the initial signal, lifting the cap above that level, after which the complementarity takes over and parents seek the evidence themselves.

3.3 The impact of the initial signal

Given the central role of the initial signal, what type of initial signals can clear the threshold? The answer is not simply those that supply more observable features that fit the alternative. Unlike in models with one-dimensional signals, how evidence is framed is central here, for two reasons: the evidence constraint rests on relative likelihoods, and a signal is a vector of observable features that each mental model reads through its own attention. Because (1) is a linear Gaussian model, each L_k is available in closed form and each Bayes factor decomposes into three interpretable channels, which turns the evidence constraint into a design instrument. Under (1), $x \sim \mathcal{N}(\mu_k b_k,\ \Sigma + \sigma_k^2 b_k b_k')$ under mental model k, and $L_k(x)$ is that density evaluated at x (Lemma OA.2 in the Online Appendix).

Proposition 3 (Three-channel decomposition of the evidence constraint)**.** *Define, for each mental model* $k = 0, 1, \ldots, K$*,*

$$\underbrace{R_k := x'\Sigma^{-1}x - s_k^2/\nu_k^2}_{\textit{off-attention residual}}, \qquad \underbrace{q_k := \frac{(s_k-\mu_k)^2}{\sigma_k^2+\nu_k^2}}_{\textit{standardized surprise}}, \qquad \underbrace{c_k := \log\left(1+\frac{\sigma_k^2}{\nu_k^2}\right)}_{\textit{Occam factor}}.$$

Then $R_k \geq 0$ *and, for each alternative k,*

$$\log \Lambda_k(x) \;=\; \tfrac{1}{2}\Big[\underbrace{(R_0 - R_k)}_{\textit{encoding}} + \underbrace{(q_0 - q_k)}_{\textit{surprise}} + \underbrace{(c_0 - c_k)}_{\textit{Occam}}\Big]. \tag{15}$$

Consequently:

- *(i) The encoding channel vanishes identically if and only if* $b_0 \propto b_k$*.*
- *(ii) The surprise channel is the only channel that depends on the prior means: it is positive if and only if the initial signal, read through each model's attention mapping, lies further from the default's prior mean than from the alternative's in standardized units,* $q_0 > q_k$*.*
- *(iii) The Occam channel is strictly negative whenever* $\sigma_k^2/\nu_k^2 > \sigma_0^2/\nu_0^2$*, in particular whenever the alternative has the default's loading,* $b_k = b_0$*, and the more diffuse prior.*

$\overline{W}_k(x)$ *is increasing in each of the three channel differences of alternative k, decreasing in each channel difference of every other alternative, and equals the Bayesian posterior model probability under a uniform prior over mental models.*

The decomposition identifies three margins on which a signal can raise the evidence cap of an alternative. The first is *encoding*, the channel $R_0 - R_k$: the signal carries features that the alternative reads but the default ignores. The second is *surprise*, the channel $q_0 - q_k$: the alternative is less surprised by the part of the signal it reads than the default is by the part it reads, as when it better predicts an observation that is extreme on a

feature both mental models read. The third is *specificity*, the Occam channel $c_0 - c_k$. A marginal likelihood penalizes a model for spreading predictive mass, and c_k is the Occam factor of Bayesian model comparison (MacKay, 1992; Jefferys and Berger, 1992; Kass and Raftery, 1995; Ba, 2026). Every alternative therefore begins at a handicap relative to the more precise default, and the handicap shrinks as the alternative's prior approaches the default's in precision. Since c_k is increasing in σ_k^2, an optimistic but vague alternative ("the program might really help") is harder to justify than a specific one ("students' scores in agriculture rise by about eight points"), even when the vague version has the higher mean. Precision has two limits. A narrower alternative whose predictions miss loses more likelihood than a vague one. And precision lowers value: a more precise prior leaves less to learn, so $\psi_k(n)$ and with it the value gap ΔU_k fall. Narrowing an alternative therefore raises its evidence cap but can weaken the value condition, and it helps most when the cap binds.

The three-channel decomposition suggests two implications for designing information interventions via the initial signal. First, exploration can be stimulated while holding constant the observable features the default reads, by presenting features that only an alternative reads (encoding) or by stating the alternative as a specific, checkable claim consistent with what is observed (specificity). An intervention can thus activate an alternative without adding to the evidence the default already reads. The intervention of Hanna et al. (2014) can be read in these terms. Seaweed farmers with years of experience had not learned that pod size mattered for yield, although the relationship was present in their own plots. A summary of their own trial data, organized by pod size, then changed their practices although it contained no observation they had not already made; in the present framework, the summary is a signal that an alternative reads and the default does not. Second, a signal can fail to change behavior not for lack of observable features but for lack of any mental model that reads or explains them: a feature no mental model on the menu reads leaves every channel, and hence every evidence cap, unchanged. Evidence alone may therefore not be enough, and an intervention may first need to supply a mental model that reads it. In the school-based agricultural education program of Lee (2026), parents are used to viewing farm work and schooling as substitutes, and an innovative development program designed to make them complements, in which students' farming builds their skills and spreads techniques to their households, has no reading on that menu; the videos and field days supply one before any evidence on the program can register.

3.4 A formal definition of curiosity

The results so far characterize when the DM entertains a new perspective and how much information she then acquires. Curiosity—the drive to seek the information required to explore new perspectives—requires no additional modeling: it emerges from the setup. Once an alternative carries weight, the DM must choose one act for two mental models that recommend different ones, and the value she loses to that compromise, the hedging cost, is by construction what she would gain from learning which model is right. A drive to resolve model uncertainty via information acquisition is therefore already present in the objective. This drive shares its structure with the information-gap definition of curiosity in psychology, a gap between what one knows and what one wants to know, which exists only once a question has been posed (Loewenstein, 1994). While Loewenstein (1994) gives the information gap a perceptual explanation—a cognitively induced deprivation that arises from perceiving the gap—this paper models the motive to close it as instrumental: closing the gap is valuable because it changes what the DM does. Two reasons support this choice. First, many non-instrumental motives can be read from an instrumental perspective. A rational analysis of curiosity in psychology treats a person as exploring what most improves the usefulness of her knowledge, and on that basis reconciles the novelty- and complexity-based accounts of what people are curious about (Dubey and Griffiths, 2020). There is also experimental evidence suggesting that curiosity responds to the stakes and the future usefulness of what is to be learned (Kobayashi et al., 2019; Liquin and Lombrozo, 2020). Second, and more importantly, the instrumental motive is the one an expected-utility maximizer already has: she values information exactly to the extent that it improves her decision (Blackwell, 1953). A choice-theoretic baseline built on it depends on no

individual differences in tastes. This lets the paper integrate non-instrumental motives on top of it (Section 3.5) and offer a unified framework. Here the question is posed by activation, the gap is the disagreement between the mental models held, and its pull is the hedging cost, which accordingly serves as the definition.

Definition 1 (Curiosity)**.** At weights $\mathbf{w}$, *curiosity* is the value of resolving which mental model is right, which is the hedging cost (7),

$$A(\mathbf{w}) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(m)}{2\lambda},$$

the component of the value of information that comes from discriminating between the models rather than from refining either.

Curiosity is thus a function of the weights, determined jointly with the information the DM acquires, and it shows in behavior through that information: how much and along which features. Three properties follow. Curiosity is zero under the default alone, however diffuse that model is. When a single alternative k is held it equals $w_k(1-w_k)D_k$, inverted-U in the weight and greatest when the DM is least sure which model applies, the intensity profile of the information-gap account, in which curiosity peaks at intermediate confidence (Loewenstein, 1994; Kang et al., 2009). And it values only signals that discriminate between the mental models, not signals that refine both, so it is directed at the observations that would settle which model is right. The last property separates curiosity from the value of information as such: the value of refining either mental model is available under a single model and is not curiosity in the sense used here.[13]

The next result completes the case for Definition 1. Suppose the act is evaluated at the weights updated by (6). Let $\bar{m}_n := \sum_{k=0}^{K} w_{k,n}\, m_{k,n}$ be the mixture posterior mean after the signals, and let $A_n := \mathrm{Var}_{\mathbf{w}_n}(m_{\cdot,n})/(2\lambda)$ be the hedging cost the DM faces at step (4), the analogue of (11) at the updated weights and means.

Proposition 4 (The value of information with updating)**.** *Under* (8) *with φ linear and the update* (6)*, the ex-ante value to the DM of the n signals, evaluated under the mixture* $\mathbf{w}$*, is*

$$\mathrm{VOI}(\mathbf{w},n) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(\bar{m}_n)}{2\lambda} \;=\; \underbrace{\frac{\sum_{k=0}^{K} w_k\, \psi_k(n)}{2\lambda}}_{\textit{within-model refinement}} \;+\; \underbrace{A(\mathbf{w}) - \mathbb{E}_{\mathbf{w}}[A_n]}_{\textit{resolution of model uncertainty}} \quad . \tag{16}$$

(a) The resolution component vanishes whenever the DM holds a single mental model, $w_k = 1$ for some k, and converges to the hedging cost, $A(\mathbf{w}) - \mathbb{E}_{\mathbf{w}}[A_n] \to A(\mathbf{w})$ as $n \to \infty$: in the limit, information recovers exactly the cost of holding several mental models at once.

(b) Let a single alternative k be held against the default, with $w_k \in (0,1)$, and let θ_0 denote the parameter generating the subsequent signals. The following limits hold almost surely as $n \to \infty$.

(i) (Shared signals.) *If both mental models read the signals as informative about θ, then $m_{k,n} - m_{0,n} \to 0$ and $w_{k,n} \to w_k\Lambda_k^\infty/(1-w_k+w_k\Lambda_k^\infty) \in (0,1)$, where $\Lambda_k^\infty := g_k(\theta_0)/g_0(\theta_0)$ is the ratio of the two models' posterior densities of θ at θ_0 after the initial signal.*

(ii) (Alternative-specific signals.) *Under Assumption 1, if the signals are generated as the alternative says, then $w_{k,n} \to 1$; if they are generated by the density f_0 the default assigns them and f_0 lies outside the alternative's family, then $w_{k,n} \to 0$.*

Part (a) makes precise what the introduction calls resolving the model uncertainty that activation creates: the hedging cost is by construction the value of learning which mental model is right, and enough signals

[13] Given the weights, the value of information here is an expected-utility object: the value of refining the mixture belief plus the expected value of perfect information about which model is right. What is not an expected-utility object is activation, the choice of the weights, and it is activation that makes curiosity switch on, scale with the weight, and be absent under the default. A Bayesian who holds two models is always curious; the DM here is curious only once she has chosen to hold a second one.

deliver that value in full. Part (b) separates two things "resolving model uncertainty" can mean. With shared signals the models come to recommend the same action, so the hedging cost disappears, but the weight settles strictly inside the unit interval: the question of which model is right is never settled. With alternative-specific signals one model wins, the weight goes to 0 or 1, and the DM keeps one mental model.

3.5 Ambiguity aversion

The framework admits the non-instrumental motive discussed in Section 3.4 without a new primitive. A strictly concave φ, ambiguity aversion in the smooth ambiguity model (Klibanoff et al., 2005), makes holding two mental models costly in itself, and so makes resolving which is right valuable even when the answer would not change the action. This subsection shows that adding it changes both whether an alternative is activated and what the DM is curious about, and in a way that strengthens rather than alters the baseline results. Both comparisons below are made at the time of choice, so neither relies on the dynamic consistency of the smooth ambiguity criterion (Klibanoff et al., 2009).

Proposition 5 (Ambiguity aversion and activation)**.** *Let φ be strictly concave, let $\bar{U} := (1-w_k)U_0 + w_k U_k$, and let $\xi := -\varphi''(\bar{U})/\varphi'(\bar{U})$. To second order in ΔU_k,*

$$(1-w_k)\varphi(U_0) + w_k\varphi(U_k) \;=\; \varphi(\bar{U}) - \tfrac{\xi}{2}\,\varphi'(\bar{U})\,w_k(1-w_k)(\Delta U_k)^2,$$

so holding alternative k carries an ambiguity premium $\frac{\xi}{2}\varphi'(\bar{U})\,w_k(1-w_k)(\Delta U_k)^2$ *beyond the hedging cost. Proposition 1 holds with D_k replaced by $\tilde{D}_k := D_k + \frac{\xi}{2}(\Delta U_k)^2$: the activation thresholds rise, and the all-or-nothing regime obtains on a set of parameters that grows with ξ.*

Ambiguity aversion thus makes alternatives harder to activate, and abrupt shifts more likely, the larger the stakes $(\Delta U_k)^2$. The premium also adds a non-instrumental motive for curiosity. It is positive even when the two mental models recommend the same action, so that $D_k = 0$ and resolving which is right has no instrumental value. An ambiguity-averse DM therefore values resolving the question for its own sake, in proportion to the stakes $(\Delta U_k)^2$. This is the non-instrumental motive of the information-gap account (Loewenstein, 1994; Golman and Loewenstein, 2018), confined here to questions that bear on payoffs. The $w_k(1-w_k)$ shape shared by the hedging cost and the premium is the one that the source function of Abdellaoui et al. (2011) and Baillon et al. (2018) attaches to perceived ambiguity, so interior weights, rather than a diffuse single prior, are the object those indices pick up.

Because the premium grows with $(\Delta U_k)^2$, both effects are largest for questions with high stakes: such questions are harder to open, since the premium raises the activation threshold, but once open they are pursued more actively. Where stakes are high, interventions that activate an alternative therefore become more effective relative to those that lower the cost of information. For example, when farmers suspect that fertilizer bought on credit may be fake (Bold et al., 2017; Michelson et al., 2021), the model predicts that side-by-side plots grown from the same bag or nutrient tests at the point of sale do more than cheaper credit, and that small trial packs, by lowering the stakes, make the question easier to open. The comparison across stakes requires that absolute ambiguity aversion ξ not fall in proportion to them; if relative ambiguity aversion is constant instead, the premium and the value gap scale together.

3.6 Endogenous exposure: where the decision maker seeks information

So far the DM has chosen only how many signals to acquire, and under Assumption 1, which facilitates closed-form expressions, those signals bear on the alternative alone. Curiosity also manifests in the type of information the DM seeks once her exposure to signals is made endogenous. This subsection relaxes Assumption 1 and allows signals to bear on the default as well as on the alternative. From here on, each signal also has a direction d, $y = d'\tilde{x} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \nu_y^2)$, so that the DM chooses what information to look at in

addition to how much to acquire. Directions may include those the default reads. She chooses the direction first and then how many observations to make. A school visit that looks at crop yields is one d; one that looks at what students can do is another. With a single alternative k held at weight $w = w_k$, write $\delta := b_k m_k - b_0 m_0$ for the difference in the two models' predictions, $\Omega_0 := \Sigma + b_0 b_0' / P_0(0)$ for the default's predictive covariance, and

$$v(w) \;:=\; \underbrace{(1-w)\,\frac{b_0}{P_0(0)} + w\,\frac{b_k}{P_k(0)}}_{\text{refinement}} \;+\; \underbrace{w(1-w)\,(m_k - m_0)\,\delta}_{\text{resolution}}.$$

Proposition 6 (Optimal direction of exposure)**.** *For an observation that is weak relative to the DM's prior uncertainty, with predictive variances common to the two models, the value of an observation along d is proportional to $(d'v(w))^2/(d'\Omega_0 d + \nu_y^2)$. Among directions normalized by $d'\Omega_0 d = 1$, the DM therefore looks along*

$$d^o(w) \;\propto\; \Omega_0^{-1}\, v(w).$$

(i) At $w = 0$ she looks through the default's lens, $\Omega_0^{-1} b_0$, and at $w = 1$ through the alternative's, $\Omega_0^{-1} b_k$.

(ii) In between, the pull toward the direction of disagreement $d^* \propto \Omega_0^{-1}\delta$, the direction that most reduces the DM's model uncertainty, $w(1-w)$, is proportional to $w(1-w)(m_k - m_0)$: it is strongest when model uncertainty is greatest and when the answer would change what she does, and it vanishes when the two models recommend the same action, $m_k = m_0$, however much their predictions differ.

The proposition turns the choice of what to look at into a tradeoff between learning more within the mental models she holds (refinement) and learning which of them is right (resolution), and the weight on the alternative sets the terms. When the alternative carries little weight, she looks where the default looks, and little, because the precise default has little left to learn and there is little model uncertainty to resolve: seeing what one expects to see is the optimal use of attention for someone with little model uncertainty. The value of an observation rises with the weight on the vaguer alternative, through both terms, which is the complementarity of weights and information in another form. As model uncertainty grows, she is pulled toward the features on which the mental models disagree, and once the alternative dominates, she looks through its lens to learn how to act on it. The resolution term is curiosity at work. It vanishes under the default alone and when both mental models recommend the same action, so curiosity targets the disagreements that matter for what she does, and it mirrors the directedness of curiosity in the information-gap account (Loewenstein, 1994): the gap is the predictive disagreement between the mental models held. Information gathered through the default's lens at low weight is routine refinement, not curiosity; what is absent under the default alone is information about which mental model is right.

Return to the farmer who sees a neighbor's high yield. Her default attributes it to favorable rainfall; an alternative attributes it to a new farming practice. If she credits the rainfall, she watches the weather: rainfall data refine the model she holds and are absorbed by it, and nothing she sees there bears on whether the practice matters. Once she suspects the practice, she compares her neighbor's plots with her own, since the comparison is the observation on which the two explanations disagree and the one that would change what she does. Once she has adopted the practice, she stops comparing and studies how to apply it on her own farm, refining the alternative as she once refined the default. Likewise, for the school-based agricultural education program in Section 3.2: a parent who takes the school farm for a typical school garden only observes that students are farming; one who starts to view it as an educational program observes students' management capabilities; and one who views it as an integrative program of both agricultural extension and education assigns plots for students to try new farming practices.

Ambiguity aversion and the direction of search. The ambiguity premium of Proposition 5 also changes where the DM looks, since a signal changes the premium through the weight and the value gap it leaves behind.

Proposition 7 (Ambiguity aversion tilts search toward discriminating signals)**.** *Under the second-order expansion, the value of n signals to an ambiguity-averse DM equals its value under linear φ plus*

$$\tfrac{\xi}{2}\,\varphi'(\bar{U})\Big[(\Delta U_k)^2\, w_k(1-w_k)\;-\;\mathbb{E}\big[(\Delta U_{k,n})^2\, w_{k,n}(1-w_{k,n})\big]\Big].$$

(i) A signal that moves only the weight, one that discriminates between the models without refining either, makes this term nonnegative, because $w_{k,n}$ is a martingale and $w(1-w)$ is concave.

(ii) A signal that moves only the posterior means, one that refines both models without discriminating between them, makes it nonpositive, because $\Delta U_{k,n}$ is a mean-preserving spread of ΔU_k. Ambiguity aversion therefore raises the value of discriminating signals and lowers the value of refining ones, and the tilt is increasing in ξ.

An ambiguity-averse DM thus seeks more information along the direction of disagreement of Proposition 6 and less about outcome data both mental models can absorb than a linear DM with the same beliefs would. The smooth ambiguity aggregator, which plays no role in the closed forms, is what sets the *direction* of search: the more the DM dislikes model uncertainty, the more her information seeking is aimed at ending it rather than at refining the mental model she favors.

4 Behavioral Effects

Standard decision problems take the DM's representation of the situation as given. Associative-memory models, building on the psychology of memory (Howard and Kahana, 2002; Kahana, 2012), instead make that representation an outcome of recall: what comes to mind depends on its similarity to the situation, the cues present, and competition from other memories, and beliefs and choices follow from what is retrieved. One group of studies shows how recall shapes beliefs: people judge probabilities by the scenarios that come to mind (Gennaioli and Shleifer, 2010; Bordalo et al., 2023), imagine the future by drawing on memory (Bordalo et al., 2025), and react to news through cue-driven recall (Enke et al., 2024). Another shows how recall shapes choice: recalled experiences set the norm against which an option is judged (Bordalo et al., 2020), and a problem is recognized by retrieving the most similar category of past problems (Bordalo et al., 2026). Framing and context thus matter because they change what comes to mind, and so which problem the DM believes she is solving.

The framework here shares that premise. Associative memory supplies the default and the menu, and the mental model in use determines which features the DM attends to and how she reads them. What, then, do we gain from modeling the DM's capacity to be curious, on top of a default classification based on associative memory? Because she can seek information and reason beyond what associative memory supplies, the model revises two hallmark behavioral effects of associative-memory models, context effects and interference effects. This section illustrates how both are revised.

4.1 Context effects

In psychology, a context effect is the finding that cues present when an experience is stored and when it is recalled shape what is remembered, even when the cues are irrelevant to the task (Tulving and Thomson, 1973; Godden and Baddeley, 1975; Smith and Vela, 2001); the temporal context model formalizes how context drives retrieval (Howard and Kahana, 2002; Kahana, 2012). Economic associative-memory models build on this finding: a payoff-irrelevant cue changes which experiences or representations come to mind, and so changes beliefs and choices (Gennaioli and Shleifer, 2010; Bordalo et al., 2020; Enke et al., 2024; Bordalo et al., 2026). For example, a consumer sees a jar of jam in plain packaging as a routine purchase and focuses on its price, but the same jar in fancy packaging becomes a treat and draws her attention to its quality.

Here a contextual cue, such as where, when, or by whom a situation is presented, is a component of the initial signal x like any other: it is evidence only under a mental model that reads it. The encoding channel of Proposition 3 determines when context matters. Call a component ζ of the initial signal *payoff-irrelevant under the default* if the default does not read it, $b_0'\Sigma^{-1}\zeta = 0$.

Corollary 2 (Context effects)**.** *Let x and $x+\zeta$ be two observations, where ζ is payoff-irrelevant under the default.*

(i) If no mental model on the menu reads ζ, that is, $b_k'\Sigma^{-1}\zeta = 0$ for every k, then $\Lambda_k(x+\zeta) = \Lambda_k(x)$ for every k, so the evidence constraints, the weights, the information acquired, and the act are all unchanged.

(ii) Otherwise ζ moves Λ_k only for the alternatives k that read it, through their statistics s_k, and moves the bound of any other alternative only through the denominator of (5)*.*

The proof is immediate from (15): $x'\Sigma^{-1}x$ cancels from $R_0 - R_k = s_k^2/v_k^2 - s_0^2/v_0^2$, so every term of $\log\Lambda_k$ depends on x only through the statistics $s_j = b_j'\Sigma^{-1}x/(b_j'\Sigma^{-1}b_j)$, and a component no model reads leaves each of them unchanged. Part (ii) is the formal content of the encoding margin. Because the default does not read ζ, its statistic s_0, and with it the default's surprise q_0, is unchanged, and the Occam channel does not depend on the initial signal at all, so ζ can act only through the statistic s_k of an alternative that reads it. In associative-memory models, adding a payoff-irrelevant cue changes which representation of the situation comes to mind and so changes behavior (Bordalo et al., 2026). Here a cue sways behavior only if it is payoff-relevant under some mental model on the menu and lifts that model over the activation threshold of Proposition 1; if it does, the curiosity this triggers tests the shift, as the next result shows.

Corollary 3 (Exposure and the path of the shift)**.** *Let observations along the direction she chooses be generated by a density f, and let f_k and f_0 be the predictive densities of alternative k and the default along it. Each observation changes* $\log\Lambda_k$ *in expectation by* $\mathrm{KL}(f\,\|\,f_0) - \mathrm{KL}(f\,\|\,f_k)$*: the shift continues on the DM's own exposure if the alternative fits the observations better than the default, and reverses otherwise. If $f = f_k$, the change is* $\mathrm{KL}(f_k\,\|\,f_0)$*, positive whenever the two predictive distributions differ; if $f = f_0$, it is* $-\mathrm{KL}(f_0\,\|\,f_k)$*. At $w_k = 0$ no information about the alternative is sought and neither occurs.*

This delivers persistence from choices rather than from a state variable: a shift that crosses the threshold carries itself forward when the alternative fits the observations better than the default and corrects itself otherwise, because the DM now looks where the answer lies, while below the threshold her exposure stays governed by the default and nothing she sees moves the bound (Figure 3).

Remark 1 (Misdirected search). When one of the two mental models is true, reversal of a mistaken shift in Corollary 3 does not require the DM to look along d^*. If the default is true, then for any direction d an observation changes the log Bayes factor of the alternative, in expectation, by $-\mathrm{KL}(f_0\,\|\,f_k)$ along d, which is nonpositive and strictly negative whenever the two predictive distributions differ; and by Lemma OA.1(i) the odds of the alternative against the default cannot exceed its updated Bayes factor. The argument holds for any rule that selects d, optimal or not and possibly history-dependent, so it covers departures from the optimal direction of Proposition 6, such as a tendency to seek confirming evidence, that is, to favor directions in which the alternative predicts a confirming outcome. Such a distortion therefore slows correction, and stalls it only along directions on which the models make identical predictions, but cannot reverse its direction. The robustness requires that one of the two mental models be true. If neither is, the sign of $\mathrm{KL}(f\,\|\,f_0) - \mathrm{KL}(f\,\|\,f_k)$ can depend on d, and a distortion of where the DM looks can select directions along which the alternative fits better, even though the default fits better along others, and so sustain the shift. Entrenching a shift therefore requires evidence that is biased toward it, or a menu on which no mental model is true combined with misdirected search.

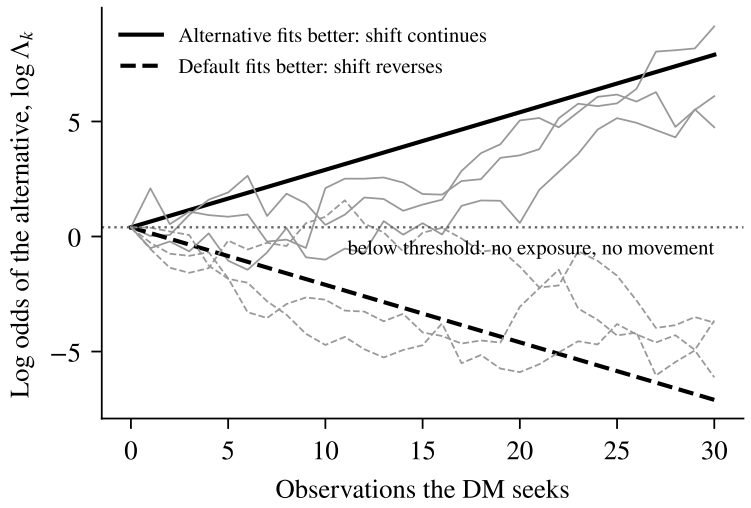


FIGURE 3. THE PATH OF A SHIFT UNDER ENDOGENOUS EXPOSURE

Notes: The figure plots the log odds of an activated alternative against the number of observations the DM seeks along her chosen direction (Corollary 3), with Gaussian predictive densities whose means differ by 0.5 standard deviations, so that each observation moves the log odds by ± 0.25 in expectation; thick lines are expected paths and gray lines are simulated ones. Below the threshold the DM seeks no such observations and the log odds stay put.

Corollary 2 separates two questions that associative-memory models answer together. Whether context moves behavior depends on the menu of mental models: some model must read the cue. Whether the movement lasts depends on the information environment and the menu. By Corollary 3 and Remark 1, a context effect that persists points to evidence that is scarce, too costly to seek given the weight the alternative carries (Corollary 1), or biased toward the mistaken model, or to a menu on which no mental model is true. Because the DM has an endogenous drive to seek this evidence herself, policy can make it accessible instead of delivering it: for farmers who distrust fertilizer that looks clumped but meets nutrient standards (Michelson et al., 2021), the remedy is quality testing they can obtain on their own.

4.2 Interference effects

In psychology, interference is the finding that similar memories compete at retrieval, so that recalling one item makes related items harder to recall (Underwood, 1957; Anderson and Neely, 1996; Anderson, 2003; Kahana, 2012). In economic associative-memory models, different experiences compete for retrieval, so recalling one inhibits recall of others and a prime can crowd out better information (Bordalo et al., 2023, 2025). For example, in Amazon's early days, people who thought of bookstores saw it as a firm in a mature sector: these frequent experiences crowded out the fewer but more relevant experiences of disruptive technologies, which would have suggested a future giant (Bordalo et al., 2025). In these models, the representation with the greatest cue similarity is the one that comes to mind.

By contrast, in this framework the menu delivers a result that no comparison of one alternative with the default can. When each of multiple alternatives is worth more than the default, none may be activated, because the same evidence must justify each of them and justifies none enough: the evidence cap of each alternative is deflated by every other alternative that looks plausible, since the automatic matching of cues against stored mental models weighs all of them at once (Section 2.3), the DM's attention is rivalrous across them, and the superadditive cost of holding several priors at once raises the price of search. Two forces make a menu of alternatives harder to act on than a single one: the cost function penalizes spreading attention (Lemma 2), and the evidence caps crowd one another. These are interference effects, but not the kind associative-memory models study, where the representation with the greatest cue similarity comes to mind. Here every alternative

remains available, and rivals interfere by dividing the support the evidence provides. What interferes is thus a feature of the information environment, such as a cue that lends plausibility to a rival or evidence spread across several alternatives rather than concentrated on one, acting on a given menu, not a failure of retrieval.

Proposition 8 (Fragmentation)**.** *Fix x and let $\overline{W}_k$ be given by* (5)*.*

(i) (Crowding.) *For each $k \geq 1$, $\overline{W}_k$ is strictly decreasing in L_j for every $j \neq k$ with $j \geq 1$. In particular $\overline{W}_k < L_k/(L_0 + L_k)$, the bound that would obtain were k the only alternative on the menu.*

(ii) (Breadth dilutes.) *Let a treatment scale the likelihoods of a set $J \subseteq \{1, \ldots, K\}$ by a common factor e^{β}, $\beta > 0$. For $k \in J$, the resulting bound is strictly smaller when $|J| \geq 2$ than when $J = \{k\}$.*

(iii) (Joint failure.) *Consequently, in the all-or-nothing regime, a feature vector that raises several L_k moderately can leave every $\overline{W}_k$ below the level alternative k needs in Proposition 1, in which case $w_k^* = 0$ for all $k \geq 1$ and, under Assumption 1, $n^* = 0$. In the graded regime crowding caps each weight at $\overline{W}_k$, and no information is acquired once every $\overline{W}_k$ falls below the level in Corollary 1.*

Corollary 4 (Crowding by unpromising rivals)**.** *Fix x and let alternative j satisfy $\Delta U_j \leq \min\{D_j, \gamma/2\}$, so that by Proposition 1 it is never activated, whatever its bound. Without the evidence constraint, adding j to the menu leaves $(\mathbf{w}^*, n^*)$ unchanged. With it, adding j multiplies every other bound by $(1 + \sum_{i \neq j} \Lambda_i)/(1 + \sum_i \Lambda_i) < 1$, so it can prevent the activation of an alternative k that would otherwise be activated and, under Assumption 1, set $n^* = 0$.*

Without the constraint, the weight on an alternative depends on its value gap and not on any likelihood, and ΔU_j lies below the threshold of both regimes, so j receives no weight and the problem for the other alternatives is unchanged. With the constraint, j still receives no weight, but it enters the denominator of (5) through Λ_j alone, whatever its value. The DM can ignore an unpromising explanation but not the evidence for it: a plausible story she would never entertain still crowds out one she would. Under optimal expectations such an explanation could simply be dropped from the menu; here it cannot, and adding an alternative that is never chosen can change the choice. The same holds for cues: a contextual cue that such an alternative reads but the promising one does not raises the former's likelihood and so, through the denominator of (5), lowers the promising alternative's bound (Corollary 2(ii)), although the alternative it favors is never entertained.

The signal's diagnostic power, large in aggregate, is fragmented across the menu, and the default remains defensible: fragmentation can produce a curiosity failure even when every alternative outperforms the default. Where crowding lowers the bounds without pushing them below threshold, alternatives are activated at smaller weights; since the information a DM seeks rises with the weight she places on an alternative (Proposition 2), search is weaker and the shift slower, so the joint failure of part (iii) is the limit of a graded effect rather than a knife edge. By the same mechanism, curiosity and the information acquired are non-monotone in the number of rivals: zero with none, positive with one that is activated, and tending to zero as rivals are added, with no information acquired beyond finitely many (Corollary OA.2 in the Online Appendix).

Interference effects have two direct applications to the design of information interventions.

Application 1: Targeting versus breadth. Consider an information intervention with a fixed amount of evidence to distribute across alternatives, so that each alternative's likelihood rises with the share it receives. Concentrating the budget on the most likely alternative drives its evidence cap to one as the budget grows, whereas spreading it uniformly leaves every cap below $\max_k L_k / \sum_{j \geq 1} L_j < 1$ at every budget, however large (Proposition OA.1 in the Online Appendix). Breadth fails not because it delivers less information but because spreading it raises every likelihood in the denominator of every bound, so the ratios barely move. Figure 4 shows the two paths for $K = 4$: the targeted bound passes 0.9 at a budget of about four, while the uniform bound stays below 0.29 at any budget. Enumerating many reasons a program might work therefore makes several alternatives moderately plausible and none plausible enough.

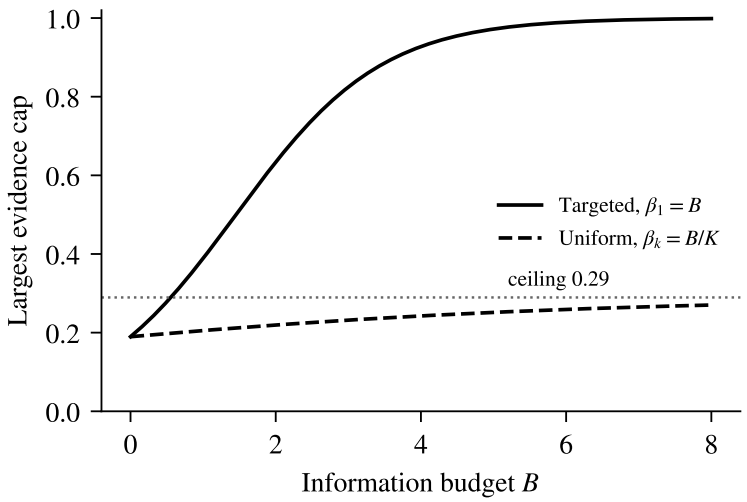


FIGURE 4. TARGETING VERSUS BREADTH IN INFORMATION INTERVENTIONS

Notes: The figure plots the largest evidence cap among $K = 4$ alternatives against an information budget B that multiplies each likelihood L_k by e^{β_k}, with $(L_0; L_1, \ldots, L_4) = (1.0; 0.55, 0.50, 0.45, 0.40)$. The targeted treatment puts the whole budget on the most likely alternative ($\beta_1 = B$); the uniform treatment spreads it evenly ($\beta_k = B/K$). The dotted line is the ceiling $\max_k L_k / \sum_j L_j \approx 0.29$ that no uniform budget exceeds (Proposition OA.1 in the Online Appendix).

Application 2: Sequencing versus breadth. Suppose two alternatives must eventually be entertained. Under (3) the cost of holding both at weights w_1, w_2 is $\frac{\gamma}{2}(w_1^2 + w_2^2) + \eta w_1 w_2$, whereas two separate activation–signal–resolution cycles cost $\frac{\gamma}{2}(w_1^2 + w_2^2)$ in total: holding them simultaneously costs a stacking premium of exactly $\eta w_1 w_2$, and a larger hedging cost whenever the two models imply different posterior means, so absent discounting sequencing weakly dominates stacking (Proposition OA.2 in the Online Appendix). Sequencing also arises on its own. If one alternative clears its threshold, it is investigated first; if the evidence goes against it, every other bound rises, so a second alternative may clear its threshold in turn. If none clears its threshold, no bound moves, and the joint failure of Proposition 8(iii) persists until a new exogenous observation arrives (Corollary OA.3 in the Online Appendix). Endogenous exposure thus sharpens targeting: the DM can climb out of a fragmented menu on her own, eliminating hypotheses one at a time, but only after something has lifted one alternative over its threshold. Until then, the curiosity failure persists.

5 Creative Thinking: Compressing the Space of Mental Models

Sections 2–4 take the menu of mental models as fixed. Many practical situations, however, involve many dimensions at once: a farmer chooses among many inputs and practices whose returns depend on one another, so the combinations to consider multiply (Laajaj and Macours, 2026). A DM who represented every possibility as a separate mental model would rarely explore at all, because of the interference effect of Section 4.2: every alternative that fits the evidence deflates the evidence cap of every other, and holding several at once is superadditively costly. Interference effects thus raise a question that a fixed menu cannot answer: how does a mind with limited capacity keep its menu small enough to explore at all?

This section and the next relax the fixed menu and model creative and critical thinking as two forms of *cognitive dimension reduction*. Exploration requires describing many possible mental models with few reusable parts, and comparing few of them at a time. Creative thinking does this by compression, building new mental models from parts the DM already holds and pooling situations under common parameters; critical thinking does it by pruning, keeping chance fits off the menu. I begin by defining templates—the reusable parts of mental models.

5.1 Templates and mental models

Let ℓ index situations, such as plots, lessons, or problems, and let $\mathscr{T} = \{1, \ldots, M\}$ be a library of templates. Informally, a template is a reusable piece of explanation: an underlying cause together with the observable features it produces, such as rain, which shows up in wet soil and high yields, or pests, which show up in damaged leaves and poor yields. A farmer who credits a neighbor's harvest to both the rain and the soil has combined two templates (Behrens et al., 2018; Whittington et al., 2020).[14]

Formally, a template is a causal mechanism: a latent cause together with the edges from it to the observable features it moves, as in a structural causal model (Pearl, 2009). I work with the linear Gaussian case, in which a template is a latent factor, a tuple $\tau_t = (b_t, \mu_t, \sigma_t^2, \rho_t)$. The loading vector $b_t \in \mathbb{R}^d$ records which observable features the cause moves: the direction in feature space along which it shows up, so that for rain it loads on soil moisture and yields. The prior $\mathscr{N}(\mu_t, \sigma_t^2)$ records how strongly the cause is expected to act, and its variance is the template's vagueness; a template the DM has not yet used is centered, $\mu_t = 0$, since she has no view of how strongly it acts. The *generality* $\rho_t \in [0,1]$ sets how far that strength is invariant across situations. A *node* is a set of templates $S \subseteq \mathscr{T}$ applied to one or more situations. Its generative process in a situation ℓ is

$$x_\ell \mid \vartheta \;\sim\; \mathscr{N}\Big(\sum_{t \in S} \vartheta_{t,\ell}\, b_t,\; \Sigma\Big), \qquad \vartheta_{t,\ell} = \mu_t + \omega_t + e_{t,\ell}, \qquad \omega_t \sim \mathscr{N}(0, \rho_t \sigma_t^2), \qquad e_{t,\ell} \sim \mathscr{N}\big(0, (1-\rho_t)\sigma_t^2\big), \tag{17}$$

with all ω_t and $e_{t,\ell}$ independent of one another and of the noise. Templates contribute additively to what is observed, as causes do in a linear structural model.

Each situation's parameter has the template's prior whatever its generality, and any two situations' parameters have correlation ρ_t, so a broader claim is not also a vaguer one. A template with $\rho_t = 0$ makes no claim across situations, since its parameters are independent draws from the prior. A template with $\rho_t = 1$ claims that one parameter governs every situation the node is applied to, and the set of situations over which its parameter is shared in this way is its *reach*: the situations across which the mechanism is invariant (Peters, Bühlmann, and Meinshausen, 2016), so that it transports from one to another (Pearl and Bareinboim, 2014). Intermediate values of ρ_t express partial pooling: the DM is unsure how far the mechanism transports.

With one template and one situation, (17) reduces to the generative process (1) of Section 2.1 with $b_k = b_t$, $\mu_k = \mu_t$, and $\sigma_k^2 = \sigma_t^2$: a mental model is a node. The extension departs from the baseline in two ways only, in that a node may combine several templates and a template's parameter may be shared across situations. Everything else carries over unchanged: a node enters the evidence constraint, the cognitive and interference costs, and the objective (4) exactly as a mental model does, and its Bayes factor against another node is computed from the same marginal likelihoods. A node is thus a causal graph, as in models of causal narratives (Spiegler, 2016; Eliaz and Spiegler, 2020), assembled from reusable mechanisms.

Definition 2 (Two types of creative thinking)**.** *Vertical creative thinking* adds a template to a node, $S \mapsto S \cup \{t\}$, in a given situation. *Horizontal creative thinking* applies a node to a new situation, for given templates.

In causal terms, a vertical step adds a cause to the graph, and a horizontal step transports the graph to a new situation. Creative thinking is thus the construction of a mental model the DM has not used before from templates she already holds.[15] Figure 5 illustrates the two steps.

[14] Templates come from slower learning or from innate structure (Spelke and Kinzler, 2007), and recent work suggests what one looks like in the brain: neural populations in the hippocampus and prefrontal cortex encode what they learn on low-dimensional surfaces, or manifolds, with a shared geometry, so that a pattern learned in one setting carries over to new ones (Bernardi et al., 2020), and the hippocampal manifold expands as experience accumulates (Zhang et al., 2023).

[15] Creation starts from existing material: imagining a new experience depends on the memory system and works by recombining stored experience (Hassabis et al., 2007; Schacter and Addis, 2007; Gupta et al., 2010). The DM chooses steps rather than finished creations, since a creation cannot be ranked before it is conceived but a step can; preferences decide whether a step is attempted, surprise gives it a direction, and evidence decides whether it is admitted (Campbell, 1960). Vertical steps capture the abductive form of creative thinking, inferring a new cause from a surprise; horizontal steps capture its analogical form, carrying a mental model to new situations.

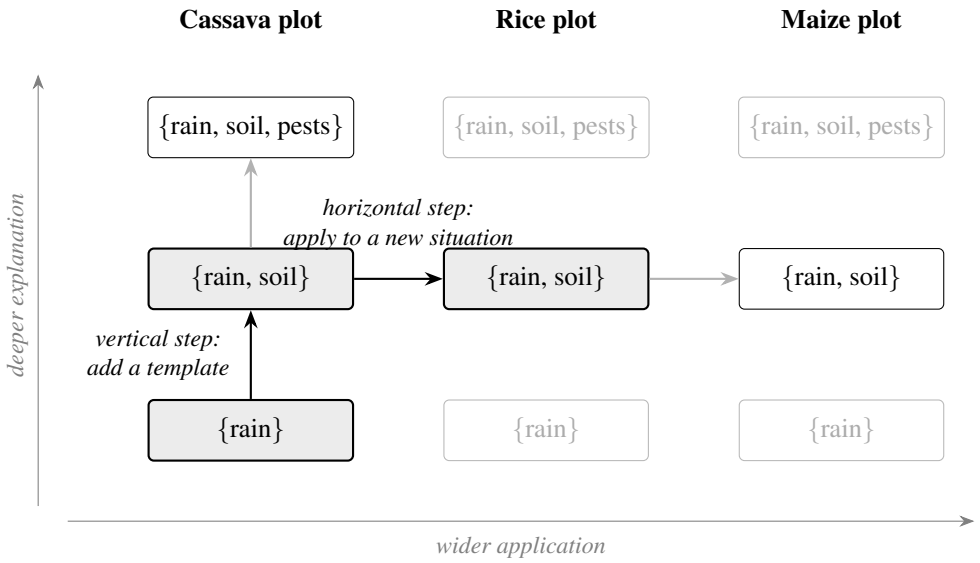


FIGURE 5. VERTICAL AND HORIZONTAL CREATIVE THINKING

Notes: Each box is a node, a set of templates applied to a situation; shaded boxes are nodes the DM holds, and gray boxes are nodes she could construct. A vertical step adds a template to the node in a given situation, here crediting a cassava harvest to the soil as well as the rain, and is admitted when the residual is surprising enough along the soil template's direction (Proposition 9). A horizontal step applies the node to a new situation, here the rice plot, and is admitted when the two plots depart from expectations in the same direction (Proposition 10). If the soil template is invariant, with $\rho = 1$, one parameter governs every plot in its reach, so a vertical step that adds it extends how far the node transfers (Proposition 11).

The two steps are the knowledge counterparts of the two kinds of innovation in endogenous growth. A vertical step is a step up a quality ladder (Aghion and Howitt, 1992): it explains a situation the DM already covers more deeply. A horizontal step resembles an expanding variety (Romer, 1990): it widens the range of situations an existing mental model serves. The two steps also reduce dimension in different directions. A vertical step adds structure within a situation and is judged as a comparison between two nodes, so the evidence is not divided among every assembly the library could form. A horizontal step pools situations: in the limit it replaces a free parameter for each situation with a single shared one.

Attempting a step to a node S' costs mental effort F, and the DM attempts it when the value of holding S', at the weight its evidence cap admits, covers the cost:

$$V\left(\overline{W}_{S'}(x), n^*\right) - V\left(\overline{W}_S(x), n^*\right) \;\geq\; F. \tag{18}$$

A step to S' is *admitted* if the evidence favors S' over the node it would replace, that is, if the evidence cap of S' in a comparison of the two exceeds one half.

5.2 Vertical creative thinking

Fix one situation and a node S. Write B_S for the matrix of the node's loadings, Σ_S for the diagonal matrix of their prior variances, and

$$V_S := \Sigma + B_S \Sigma_S B_S'$$

The approach is close to combinational creativity and insight as a change of representation (Koestler, 1964; Boden, 2004; Knoblich et al., 1999): a library of M templates describes up to $2^M - 1$ mental models (Weitzman, 1998), so it represents a large space with few parts. It differs from models of growing awareness, in which unforeseen options arrive from outside (Karni and Vierø, 2013). Boden (2004) distinguishes a third type, transformational creativity, which changes the space of possible ideas itself; since new structural representations are themselves learned from experience (Whittington et al., 2020), the model reads it as recombination at the level of the templates or as the product of the slower learning by which new templates form, which lies outside the model.

for the node's predictive covariance of the features. Let $r := x - B_S\mu_S$ be the part of the initial signal that node S did not predict, and let $h_t := b_t' V_S^{-1} b_t$ be the precision with which template t's direction can be read against everything node S already explains.

Proposition 9 (When a vertical step is admitted)**.** *Let $\upsilon_t := \sigma_t^2 h_t$ measure the template's vagueness relative to the precision with which its direction can be read, the counterpart of σ_k^2/v_k^2 in Proposition 3. The log Bayes factor of the node $S \cup \{t\}$ against S is*

$$\log \Lambda_{S\cup t:S} \;=\; \tfrac{1}{2}\Big[\,\frac{\upsilon_t}{1+\upsilon_t}\,z_t^2 - \log(1+\upsilon_t)\Big], \qquad z_t := \frac{b_t' V_S^{-1} r}{\sqrt{h_t}}, \tag{19}$$

where z_t is the standardized surprise of the residual along the template's direction, the surprise channel of Proposition 3 in one direction, and $\log(1+\upsilon_t)$ is the template's Occam factor. In a comparison of the two nodes, the step is admitted, $\overline{W}_{S\cup t} > 1/2$, if and only if

$$z_t^2 \;>\; \frac{1+\upsilon_t}{\upsilon_t}\log(1+\upsilon_t), \tag{20}$$

a threshold that is increasing in υ_t and tends to one as $\upsilon_t \to 0$.

Proposition 9 shows where compression pays. Because the DM evaluates a step, the comparison is between two nodes, and the test is one-dimensional: what matters is the surprise of the residual along the template's direction, z_t, net of an Occam penalty for the template's vagueness. The three margins of Section 3.3 reappear in a single direction: surprise and encoding both enter through z_t, the first through features S reads but mispredicts and the second through features S ignores, which cannot move any decision made under S but can justify a new template (Corollary OA.4 in the Online Appendix), while specificity is the Occam penalty, so a vaguer, more open template is harder to adopt, not easier. Compression has a price. If the DM tries many templates after seeing x and keeps the best, the candidates she tried form a menu in all but name, and the crowding that would have disciplined that menu is lost.

5.3 Horizontal creative thinking

Now hold the templates fixed and ask whether a node applies across N situations. Consider one template, with prior $\mathcal{N}(\mu, \sigma^2)$ and generality ρ; in a node with several templates the argument applies to each template's parameter given the others. Let s_ℓ be the template's sufficient statistic in situation ℓ, read as in Lemma 1, and measure it in units of its reading noise, so that $s_\ell \mid \vartheta_\ell \sim \mathcal{N}(\vartheta_\ell, 1)$. The *local* node sets $\rho = 0$ and treats the N parameters as unrelated draws from the prior. The *general* node shares the parameter across situations; I state the invariant case $\rho = 1$, in which one parameter governs all N situations, and leave intermediate ρ to the Online Appendix. Write $d_\ell := s_\ell - \mu$ for each situation's departure from the prior, $\bar{d} := N^{-1}\sum_\ell d_\ell$ for the departure the situations share, and $SS := \sum_\ell (d_\ell - \bar{d})^2$ for their dispersion.

Proposition 10 (When a horizontal step is admitted)**.** *At $\rho = 1$, the log Bayes factor of the general node against the local node is*

$$\log \Lambda_{\text{gen:loc}} \;=\; \underbrace{\tfrac{1}{2}\Big[N\log(1+\sigma^2) - \log(1+N\sigma^2)\Big]}_{\textit{Occam gain}} \;-\; \underbrace{\tfrac{1}{2}\,\frac{\sigma^2}{1+\sigma^2}\,SS}_{\textit{dispersion penalty}} \;+\; \underbrace{\frac{N(N-1)\,\sigma^4}{2(1+\sigma^2)(1+N\sigma^2)}\,\bar{d}^2}_{\textit{shared-departure gain}}. \tag{21}$$

It is zero when $N = 1$. For $N \geq 2$, it is increasing in the shared departure $\bar{d}^2$ and decreasing in the dispersion SS, so the general node is admitted if and only if SS falls below a threshold that is increasing in $\bar{d}^2$. The Occam gain is increasing in N.

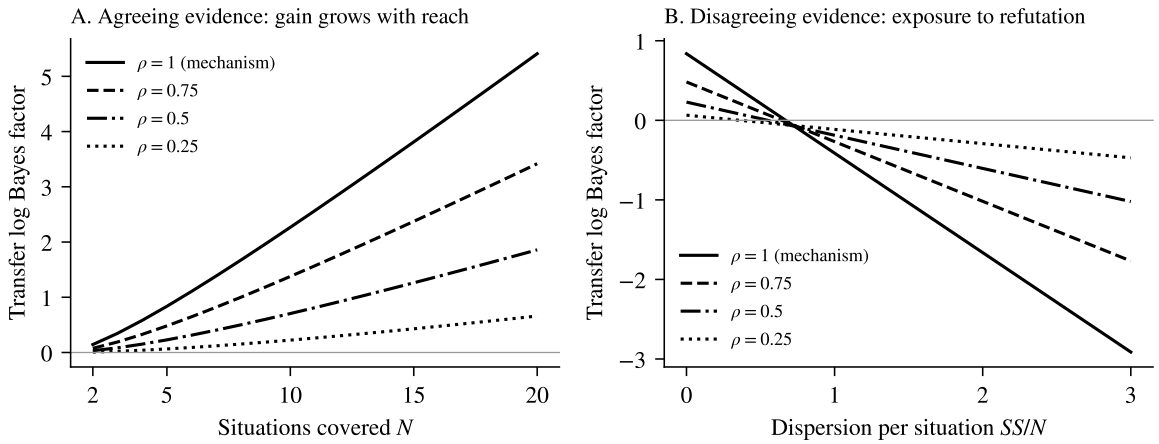


FIGURE 6. DEEPER EXPLANATIONS APPLY MORE WIDELY

Notes: Both panels plot the log Bayes factor of the general node against the local node from the general form of Proposition 10 (Online Appendix), with $v = \sigma^2 = 1$ and no shared departure ($\bar{d} = 0$), for four levels of generality ρ. Panel A varies the number of situations the template covers when their evidence agrees ($SS = 0$). Panel B fixes $N = 5$ and varies the dispersion of the evidence across situations. An invariant template ($\rho = 1$) gains most from agreement and loses most from disagreement.

For $0 < \rho < 1$ the same three terms appear with smaller Occam gain and smaller dispersion penalty, both increasing in ρ (Proposition OA.3 in the Online Appendix).

Horizontal creative thinking reduces dimension across situations. The local node carries N free parameters; the general node at $\rho = 1$ carries one, and the Occam gain from that reduction grows with N. Evidence from different situations then bears on a single parameter rather than being divided among N, the cross-situation counterpart of concentrating information on one alternative (Section 4.2). The dispersion penalty gives a reading of a classic finding on social learning: at the onset of the Indian Green Revolution, farmers learned from neighbors' yields for wheat but not for rice, whose yields were sensitive to unobserved soil characteristics, and rice growers experimented on their own land instead (Munshi, 2004). Where situations differ, the step that carries a neighbor's experience to one's own plot is not admitted, and evidence must be gathered within the situation.

5.4 Deeper explanations apply more widely

Every template is a mechanism; an *invariant template* is one whose mechanism acts with the same strength in every situation in which it operates: its generality is $\rho_t = 1$ on its reach R_t, and outside R_t it does not contribute to what is observed.

Proposition 11 (Invariant templates determine the extent of horizontal transfer)**.** *Let $S' = S \cup \{t\}$ add an invariant template t whose parameter is common to a set of situations R_t. Then:*

(i) The reach of S' is R_t: the parameter of t is shared on R_t, and outside R_t the node is as before, so the reach is a property of the template rather than a choice.

(ii) For $N \geq 2$ situations in R_t whose evidence agrees ($SS = \bar{d} = 0$), the transfer Bayes factor of Proposition 10 is strictly larger under S' than under any node with $\rho < 1$.

(iii) For situations whose evidence disagrees, the penalty on dispersion is increasing in ρ and largest at $\rho = 1$.

Invariant templates are the limit of compression: one parameter for every situation in which the mechanism operates, with a reach set by the template rather than chosen. The property that lets agreeing evidence pool (part (ii)) also makes disagreeing evidence count most heavily against the template (part (iii)). Figure 6

shows the two faces: with agreeing evidence the gain from an invariant template grows fastest in the number of situations covered, and with disagreeing evidence it falls fastest. Maximal compression is maximal exposure to refutation, which is why compression needs the discipline of the next section. In the language of growth theory, an invariant template is a general-purpose technology (Helpman and Trajtenberg, 1998): one step up the ladder raises explanatory quality in every situation where the mechanism operates, and by part (ii) it raises the return to expanding variety over the mechanism's reach. The DM chooses whether to apply a node to a new situation; whether it transfers, and how far, is decided by the templates it contains. Vertical and horizontal creative thinking are therefore not independent, yet neither substitutes for the other: the vertical step sets how far a node can transfer, and the horizontal step is still required to realize the transfer and must be justified by the evidence in each situation. This differs from models that treat the two kinds of innovation as separate activities (Aghion and Howitt, 1992; Howitt, 1999). Because creation draws only on the templates the DM holds, creative thinking is not a fixed trait but a capability that can be cultivated: providing templates, and above all invariant ones, enlarges both the set of mental models she can construct and the range of situations to which they transfer.

This matters for training teachers and agricultural extension officers. A set protocol gives them a precise local mental model, which pays a small Occam penalty and is easily activated where it was designed but carries no claim beyond it; understanding why a practice works supplies an invariant template, whose gain is zero in a single setting and grows with the number of classrooms and farms it covers (Proposition 10), at the cost of greater exposure to evidence that disagrees (Proposition 11(iii)). This qualifies a common presumption that teaching why a practice works is enough to make trainees generalize. The template does not deliver the adaptations ready-made: each is a horizontal step the trainee must take and test herself, and it happens only if her curiosity clears the threshold in the new setting (Section 3.1), so training in mechanisms pays through her continued curiosity and horizontal creative thinking, not in place of them. What makes trainees generalize is thus not the templates alone but the curiosity and horizontal creative thinking to apply them, and the opportunities that make each application pay. This complements recent work that formalizes how teaching mechanisms improves decisions in agriculture: in Davies and Sankar (2026) and Sankar et al. (2026), a mechanism is a known truth that lets a Bayesian design better signals and generalize within a single decision; here it is a hypothesis whose value is realized only across situations, and only if the horizontal step is taken and admitted.[16]

The framework also offers an interpretation of skill in agriculture. Laajaj and Macours (2026) find that high-skill farmers learn more from agronomic trials, and model skill as the precision with which a farmer reads each observation. Here skill is the library of templates: a DM who holds more of them compresses the same evidence further and needs fewer trials, and the advantage should be largest in decisions with many interacting inputs, where they find it concentrated.

6 Critical Thinking: Pruning the Menu with Fresh Evidence

Compression creates its own risk. A library of templates makes candidate mental models cheap to construct, and a DM who searches them after seeing the evidence and keeps the best fit has considered a large menu while counting only the winner. The rivals she discarded drop out of the evidence cap, and with them the crowding of Proposition 8 that a menu of that size would have faced. Critical thinking prunes the menu,

[16] Davies and Sankar (2026) show that conceptual knowledge, knowing how unknown states relate, is valuable because it lets the agent design more informative signals; the value is largest when states reduce to few concepts and vanishes as signals become abundant. Sankar et al. (2026) add mechanistic explanations to fertilizer demonstrations in Uganda and find that treated farmers generalize better across soils and obtain higher yields. Three differences from the present framework: the value of a mechanism arises through transfer across situations rather than sampling within one, so it is zero in a single situation and grows with the situations covered; it is realized only if the DM takes the horizontal step and the evidence in each situation admits it; and a mechanism can fail to be activated, be crowded out by rivals, or be refuted.

admitting only mental models that survive evidence not used to form them. That searching many explanations yields false ones is common sense; the results below say when discipline is enough, a threshold that rises with the templates' vagueness, what it buys beyond avoided errors, a higher evidence cap for the right alternative, and what it costs, a curiosity failure for correct insights.

The problem is familiar to empirical economists as multiple hypothesis testing. A researcher who tests many outcomes and reports the one that is significant finds false positives at a rate that grows with the number of tests, and flexible analysis can make almost any result look significant (Simmons et al., 2011; Gelman and Loken, 2014). Corrections for multiple inference and pre-analysis plans are now standard in field experiments for this reason (Anderson, 2008; Olken, 2015; List, Shaikh, and Xu, 2019). Critical thinking is the cognitive counterpart: a person who searches many explanations after the fact faces the same multiplicity, and the same remedies apply.

Coming up with a new explanation is only half of learning. The other half is judging whether it is right. A common failure is to pick an explanation because it fits and then count that same fit as evidence that it is right: the data are used twice. A farmer whose neighbor harvests far more than usual runs through possible reasons, such as the seed, the rain, the planting date, and the spacing, and settles on spacing because it matches what she saw most closely. With enough candidate reasons, one is bound to match by chance, so the match alone says little about whether spacing caused the harvest. A critical thinker asks whether she would have found a fitting story whatever had happened, and she checks the story on cases she did not use to find it. Reading meaning into chance coincidences is also how superstitions form (Skinner, 1948). Here, *critical thinking* is the disposition to distinguish predictions from after-the-fact explanations, so that every mental model, new or old, is judged only by evidence that was not used to form it. It is close to the disposition to evaluate arguments independently of prior belief (Stanovich and West, 1997) and opposite to confirmation bias (Nickerson, 1998). This is one component of critical thinking, the discipline against after-the-fact fitting; the analysis of arguments and the evaluation of sources lie outside the model.

6.1 The search discount

Fix a node S and an observation x. After seeing x the DM searches a class $\mathscr{S} \subseteq \mathscr{T}$ of candidate templates and keeps $t^* := \arg\max_{t \in \mathscr{S}} \Lambda_{S \cup t:S}(x)$, with the Bayes factor of Proposition 9. A Bayesian who places equal prior weight on each candidate assigns prior odds $1/|\mathscr{S}|$ to t^*.

Definition 3 (Critical thinking)**.** A DM with critical thinking $\kappa \in [0,1]$ evaluates a node constructed by searching a class $\mathscr{S}$ after observing x at the log odds

$$\log \Lambda_{S \cup t^*:S}(x) \;-\; \kappa \log |\mathscr{S}|. \tag{22}$$

A model specified before x, such as one supplied by teaching, has $|\mathscr{S}| = 1$ and is not discounted, and neither is evidence that was not used in the search.

With $\kappa = 1$ the candidates searched share equally the prior weight that a single alternative named in advance would carry, so the winner is charged for the rivals the search considered: this is the evidence constraint of Section 2.3 applied to the menu the search created, and $\kappa < 1$ is the only departure from it. The discount depends on the breadth of the search, not on its outcome, and plays the role of a multiplicity correction: an explanation named before the evidence is treated like a hypothesis registered in a pre-analysis plan, while one found by searching many candidates is held to a higher bar.

6.2 Search breadth and false insights

A *false insight* is a template admitted when the initial signal was generated by S. Let the candidates share the same $\upsilon_t = \upsilon$ of Proposition 9 and read directions that are orthogonal in the metric V_S^{-1}. Admission requires

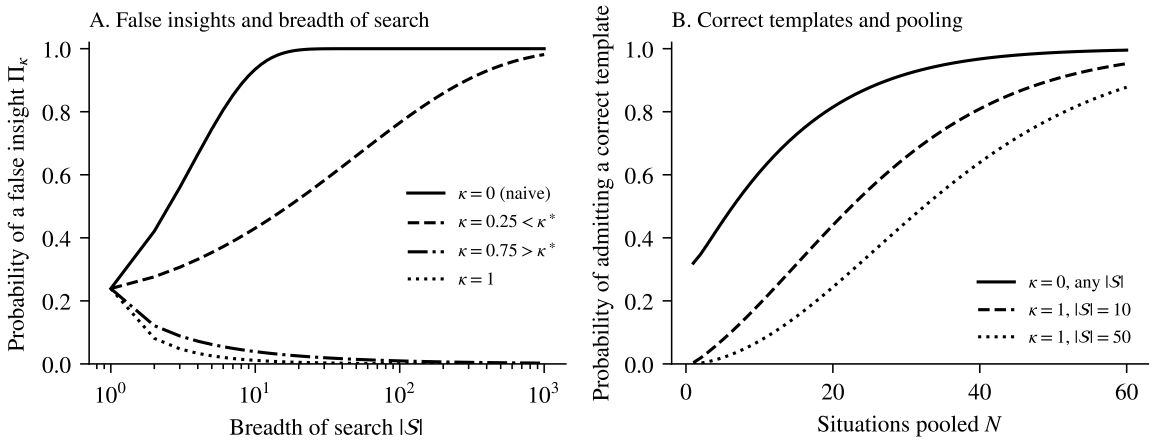


FIGURE 7. CRITICAL THINKING, BREADTH OF SEARCH, AND POOLING

Notes: Both panels set $\upsilon = 1$, so the threshold in Proposition 12 is $\kappa^* = 1/2$. Panel A plots the probability of a false insight $\Pi_\kappa(|\mathscr{S}|)$ against the breadth of search for four levels of critical thinking, with the candidates independent under S. Panel B plots the probability that a correct template is admitted against the number of situations N whose evidence bears on it (Corollary 5), with $h_t \vartheta_t^2 = 0.36$ in each situation; the discount leaves the naive DM's bar unchanged, so her curve does not depend on $|\mathscr{S}|$.

that the evidence cap of $S \cup \{t^*\}$ in its comparison with S exceed one half.

Proposition 12 (Critical thinking and breadth)**.** *Under these conditions the DM admits a template if and only if the largest surprise clears the admission bar* c_κ*,*

$$\max_{t \in \mathscr{S}} z_t^2 \;>\; c_\kappa(\upsilon, |\mathscr{S}|) \;:=\; \frac{1+\upsilon}{\upsilon}\Big[\log(1+\upsilon) + 2\kappa \log|\mathscr{S}|\Big], \tag{23}$$

and the probability of a false insight, $\Pi_\kappa(|\mathscr{S}|)$*, satisfies:*

(i) If $\kappa < \upsilon/(1+\upsilon)$*, then* $\Pi_\kappa(|\mathscr{S}|) \to 1$ *as* $|\mathscr{S}| \to \infty$*. In particular a naive DM forms a false insight with probability approaching one when she searches broadly.*

(ii) If $\kappa > \upsilon/(1+\upsilon)$*, then* $\Pi_\kappa(|\mathscr{S}|) \to 0$*, at a rate in* $|\mathscr{S}|$ *given in the Online Appendix. This part does not require independence across candidates.*

(iii) For a candidate that is right, the surprise required for admission rises with the breadth of the search, by $2\kappa \frac{1+\upsilon}{\upsilon} \log|\mathscr{S}|$ *in* z_t^2*.*

The bar in (23) rises in proportion to $\log|\mathscr{S}|$, the rate at which a Bonferroni critical value for $|\mathscr{S}|$ normal tests rises, so breadth of search and discipline in judging it go together. Panel A of Figure 7 shows the divide at κ^*: below it a broad search ends in a false insight almost surely, so partial discounting is useless, and above it almost never; since κ^* rises with υ, vaguer templates demand more discipline. Part (iii) is the seed of the potential pitfall: critical thinking also demands more of a correct insight. The discount matters beyond single steps. Each false insight a naive DM admits is a mental model that fits the evidence, so it enters the denominator of every evidence cap and dilutes the support for the alternative that is right (Proposition 8), whether or not she would ever entertain it (Corollary 4). By keeping chance fits off the menu, critical thinking can make the right alternative easier to activate. The same discount applies to old mental models: critical thinking loosens a default that owes more to after-the-fact search than the alternative does, so it is not a bias toward the familiar (Proposition OA.4 in the Online Appendix).

6.3 Fresh evidence and the pitfall of critical thinking

Fresh evidence removes the difference between the two DMs. Let $x'_1, \ldots, x'_m$ be later observations independent of x given the process that generates them, such as observations from new plots. They enter both DMs' log odds in the same way, so the critical thinker's log odds for $S \cup \{t^*\}$ against S differ from the naive DM's only by the one-time constant $\kappa \log |\mathscr{S}|$. Each fresh observation moves the log odds in expectation by how much better $S \cup \{t^*\}$ fits it than S, as in Corollary 3, and as such observations accumulate both DMs' evidence caps for $S \cup \{t^*\}$ converge to one if t^* improves the fit and to zero if it worsens it (Proposition OA.5 in the Online Appendix).

Fresh evidence plays the role of a replication or hold-out sample, and it is how the menu is pruned over time. Laajaj and Macours (2026) show what this pruning looks like in the field. After seeing agronomic trials, Kenyan farmers adopted new combinations of inputs and practices and at first made more mistakes, pairing inputs that do not work well together; the mistakes fell as they kept experimenting on their own plots. In the language here, the trial plot is the evidence that admits a combination, the farmer's own plot is the fresh evidence that tests it, and a combination that fails the test is dropped. A node whose predictions fail on observations not used to form it sees its evidence cap fall toward zero, while the discount, charged once, matters most when evidence is scarce. Extending a mental model to a new situation, the horizontal step of Section 5, is therefore also an out-of-sample test.

Fresh evidence arrives, however, only if it is collected, and the DM collects evidence only through the mental models she holds. Say that an observation *bears on* a template t if its distribution differs under $S \cup \{t\}$ and under S. The pitfall of critical thinking is therefore a curiosity failure rather than a rejection: when such observations are sought only by a DM who places weight on $S \cup \{t\}$, the discount that keeps the template off the menu also keeps the DM from collecting the evidence that would admit it, and her log odds stay where the discount left them, whether or not the template is correct (Corollary OA.5 in the Online Appendix). The higher bar of Proposition 12(iii) is temporary only when evidence bearing on the template arrives without being sought, as when a neighbor's harvest is visible anyway or a program delivers the evidence.

Creative thinking can offset the discount, but only in particular forms. The discount grows with the number of candidates searched, so generating more candidates after seeing the evidence raises it. What lowers the surprise a correct template needs is a narrower search, a template held before the evidence, or a template whose parameter is common to many situations, so that the evidence from all of them bears on it at once.

Corollary 5 (Creative thinking and the discount)**.** *Under the conditions of Proposition 12, let x contain observations from N situations that are independent given each node and share the node's predictive covariance V_S, and let t be a centered invariant template whose parameter is common to the N situations, with precision υ in each. Let $z_{t,\ell}$ be the standardized surprise along t in situation ℓ, and $Z_t := N^{-1/2} \sum_\ell z_{t,\ell}$. Then:*

(i) The critical thinker admits t if and only if $Z_t^2 > c_\kappa(N\upsilon, |\mathscr{S}|)$.

(ii) The additional surprise critical thinking requires, $c_\kappa(N\upsilon, |\mathscr{S}|) - c_0(N\upsilon, |\mathscr{S}|) = 2\kappa \frac{1+N\upsilon}{N\upsilon} \log |\mathscr{S}|$, is zero when $|\mathscr{S}| = 1$, increasing in $|\mathscr{S}|$, and decreasing in N.

(iii) If t is correct, with parameter $\vartheta_t \neq 0$, the probability that the critical thinker admits it tends to one as $N \to \infty$, for every $\kappa \in [0,1]$ and every $|\mathscr{S}|$.

Part (ii) makes the three routes precise: the discount vanishes for a template held before the evidence, falls with a narrower search, and falls as a template applies to more situations. Part (iii) shows that an invariant template applied widely enough is eventually admitted however strong the discount. Panel B of Figure 7 shows the cost and its remedy: at small N the critical thinker admits a correct template far less often than the naive DM, and pooling closes the gap. Pooling has a second effect: by Proposition 12 with precision $N\upsilon$, the threshold value of κ that keeps false insights out rises to $N\upsilon/(1+N\upsilon)$. A full discount, $\kappa = 1$, still meets it, so a DM who discounts fully and applies templates widely both keeps chance fits off the menu and admits the

correct template. Critical thinking thus raises the return to creative thinking that narrows the search or pools evidence across situations, and lowers the return to creative thinking that only adds candidates.

7 Concluding Remarks

This paper asks what stimulates curiosity: the drive to seek the information required to explore new perspectives. It treats contextual shifts in mindsets as a choice of how much model uncertainty to entertain. Associative memory supplies a familiar default. After an initial signal, a DM with latent, less precise alternatives jointly chooses how much weight to place on each and what information to acquire, and how much, disciplined by an evidence constraint that caps the weight on any alternative at its Bayesian posterior. Curiosity then arises endogenously as the value of resolving the model uncertainty the DM chooses to entertain.

Because DMs can seek information and reason beyond what associative memory supplies, exploration is both triggered and tested by evidence, which revises hallmark behavioral effects in associative-memory models. Contextual cues payoff-irrelevant under the default sway decisions only when they activate an alternative under which they are relevant, and the curiosity this stimulates tests that alternative, so the shift lasts only if evidence supports it. Yet rival explanations can stifle exploration of a promising alternative by dividing the evidence, even if the DM would never entertain them. Sections 5 and 6 relax the assumption of a fixed menu and treat creative and critical thinking as cognitive dimension reduction: creative thinking compresses the space of mental models into a few reusable templates, and critical thinking prunes the menu to those that survive evidence not used to form them. Two results depart from common sense: how far a mental model transfers is fixed by its templates rather than chosen, so a template that explains why pays only through transfer, which it does not by itself deliver; and the pitfall of critical thinking is a curiosity failure rather than a rejection.

Taken together, the results suggest that open-mindedness, curiosity, creative thinking, and critical thinking form a capability growth cycle within the brain's evidence-based reasoning (Figure 8): each process feeds the next, and each turn enlarges what the DM can explain and where. Open-mindedness—entertaining an alternative to the extent the evidence allows—is what curiosity requires: curiosity is the value of resolving model uncertainty, and there is none to resolve until an alternative is entertained. Curiosity tends to extinguish itself, since the evidence it gathers resolves the uncertainty that prompted it: the mental models come to agree or one of them prevails. Creative thinking can replenish that uncertainty. The observations curiosity seeks were not used to form the mental models, so their surprises along a template can justify a new mental model, and they are the fresh tests critical thinking requires. Critical thinking keeps chance fits off the menu, where they would dilute the evidence cap of the right alternative. Its potential pitfall ties the four together: it raises the bar for a correct template more the broader the search that found it, which creative thinking that narrows the search or pools evidence across situations offsets, and it can leave a correct template untested unless curiosity, or evidence that arrives without being sought, supplies fresh evidence. Finally, a mental model that sets the capability growth cycle going in one situation does not carry over for free: extending it to new situations is a horizontal step that the evidence must justify and that costs mental effort.

How can developing-country teachers be cultivated in new pedagogies that go beyond transmission and retention of subject content, such as inquiry-based instruction, while remaining able to tailor them to different contexts, with their different challenges, student backgrounds, and institutional constraints, and to adapt them to the changes that socioeconomic transformation brings? Teaching why a practice works does not by itself make trainees generalize, so the framework points to an integrative approach with two parts: what training targets, and the opportunities that keep the capability growth cycle going.

The first is to target templates rather than hand out mental models. This is the distinguishing feature of the "Learning to Learn" approach to teacher training (Nourani et al., 2025, 2026). Rather than lesson plans, it works through guiding texts organized around capabilities, each a developed capacity to think and act purposefully within a sphere of activity, such as critical thinking, creativity, and social responsibility, and each built from the concepts, information, skills, and attitudes it requires. Content is presented as models with a

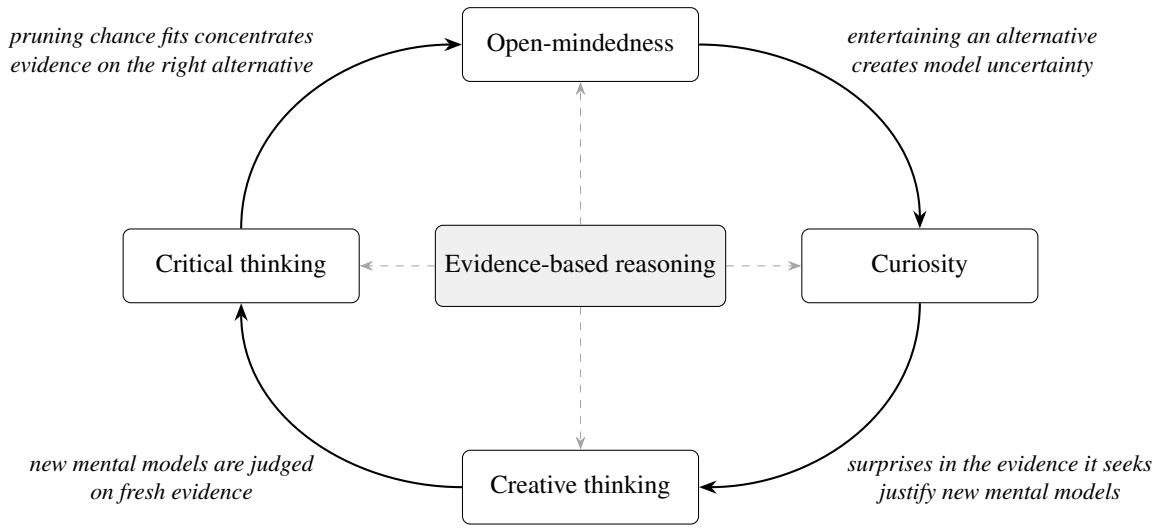


FIGURE 8. THE CAPABILITY GROWTH CYCLE

Notes: Each solid arrow summarizes a result. Entertaining an alternative within its evidence cap creates the model uncertainty that curiosity values; the surprises in the evidence curiosity seeks justify new mental models; new mental models are judged on evidence not used to form them; and keeping chance fits off the menu concentrates the evidence on the right alternative. Curiosity extinguishes itself as the evidence it gathers resolves model uncertainty, so the capability growth cycle continues only if creative thinking replenishes it. The dashed arrows indicate that every process is disciplined by the evidence constraint.

limited range of validity, and concepts are carried deliberately across domains, from the properties of matter to the qualities of a person and a community. Reflection on inquiry-based against traditional instruction sharpens these templates and directs attention to features of the classroom the traditional model ignores; simulation and discussion with peers let teachers see the mental models they assemble work, until they are confident enough to explore their consequences in their own classrooms; drawing analogies trains the horizontal steps that carry a template to new situations; and separating facts from opinions includes the discipline against after-the-fact fitting formalized here. Trained teachers can then assemble their own mental models suited to their context rather than apply one designed elsewhere.

The second part is a stream of opportunities that keeps the capability growth cycle going and lets teachers grow out of their comfort zones step by step rather than all at once. Because a shift to a new mental model must promise enough to pay for itself, the opportunities must raise the returns to the new pedagogy as well as make evidence on them accessible. School-based agricultural education supplies them as a platform that offers experiential science learning opportunities and high-return farming techniques (Lee, 2026): school farms serve as demonstration plots, science lessons and student home projects make evidence cheap and draw it from many plots at once, and extension officers return with feedback from time to time. Sharpened templates are investigated only if signals are cheap enough, a correct explanation that critical thinking holds back is tested only if evidence arrives without being sought, and an analogy that carries a template across plots clears the discount only with evidence from many situations. Each new challenge, such as a pest outbreak, a failed harvest, a new crop, or a new market, calls for a new mental model, and the stream of evidence lets teachers build and test it. A teacher can start from an inquiry routine made familiar through practice, with lower but more certain payoffs, which is easy to activate and cheap to test; testing it either rules it out, which raises the evidence caps of its rivals, or uncovers surprises that justify a richer mental model, so that mental models with lower but more certain payoffs serve as stepping stones to ones with higher but less certain payoffs, and capabilities keep growing.

My ongoing research, which restructures school-based agricultural education around the Learning to Learn approach to teacher training in a randomized trial across Liberian schools, is designed to test the value

of this integration: how much the stream of opportunities adds to the training.

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